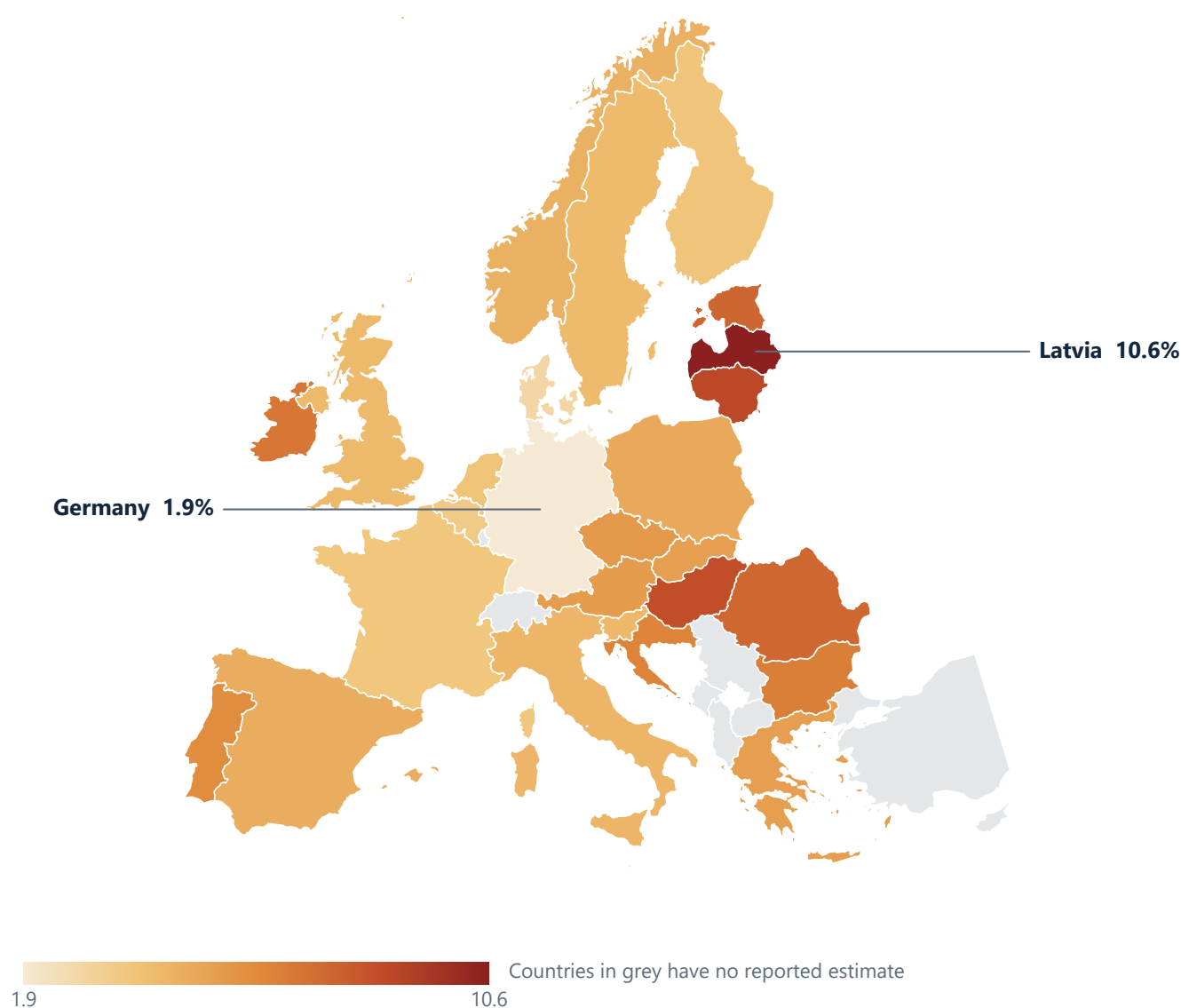


Completing Europe's power transition

Inherited capital and the financing architecture after liberalisation

Evidence from thirty countries

Cost of capital reported for a solar or onshore wind project, per cent, 2018 onward



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Completing Europe's power transition: inherited capital and the financing architecture after liberalisation

Abstract

Wind, solar and nuclear plants are paid for before they generate, so the price of money decides much of what they cost. Across Europe the price of that capital differs sharply between countries. The cost of capital reported for renewable projects runs from 1.93 per cent in Germany to 10.59 in Latvia, and an independent survey five years earlier found a similar cross-country ordering across twenty-five countries. The gap opens where money reaches the project, in how much a bank will lend and what the equity that fills the rest demands, rather than in sovereign borrowing, where the countries lay much closer together over the same years. This paper asks why a single electricity market has not produced a single market for financing the plants that supply it, and sets the question against the history of how those plants were paid for. The historical cases document five financing routes, utility balance sheets, state capital, consumer prepayment, offtaker consortia and plan allocation, drawn here from appraisals, government bills and company accounts. They differ in where the capital came from but share an institutional route through which long-lived investment could be repaid. Liberalisation separated those functions between 1995 and 2010 rather than abolishing them, and in three documented cases a country's financing regime had already changed once as its earlier fleet accumulated assets and cash flow. The instruments that followed restore parts of that arrangement and not the whole: contracts can protect a project's revenue, public lending can lower the cost of its debt, guarantees can absorb named risks and grants can cut the capital it must raise, and none of them by itself reproduces the financing capacity of an established utility with a cost-recovering customer base. Countries entered the liberalised market with different amounts of capital already paid for and different amounts left to finance on market terms. Since 2010 the building has shifted towards the countries that had built least, and the available project evidence reports several of them financing with less debt and more equity. Completing the transition takes more than mobilising a larger quantity of capital. The unresolved question is whether comparable projects can obtain comparable terms across countries when the revenue protection, the debt capacity and the risk-bearing functions once combined inside a utility are supplied separately and project by project.

Keywords: electricity, cost of capital, project finance, liberalisation, regulation, energy transition, Europe.

JEL: G31, L94, Q48, N74.

Europe's electricity system is becoming more capital-intensive. Wind, solar and nuclear plant carry **front-loaded capital expenditure**: a large share of the lifetime cost is committed before the first unit is sold, and their subsequent operating costs are relatively low beside those of a plant that burns fuel. Two things follow for the financier. Large sums have to be committed before any revenue begins, and that expenditure becomes sunk as construction proceeds, so it cannot be withdrawn if the outlook changes. What is committed is then recovered across the twenty or thirty years of operation, out of the revenue the plant earns or the support arrangements attached to it.

Those two features make the plant a **long-duration asset**, and the price of money decides what it costs. The further the repayment lies in the future, the more a given discount rate removes from its present value, so a rise in the cost of capital raises the cost of this plant far more than it raises the cost of a gas turbine that pays for much of its output as it burns fuel. How a project is financed therefore affects the return its lenders and shareholders require, and that required return raises the revenue the project needs over its life in order to recover its costs.

Before liberalisation, most European electricity systems gave investment a route to cost recovery from electricity users. That gave lenders and shareholders a route to repayment that did not depend on what an individual plant might earn in a future wholesale market. The historical cases therefore separate ownership from the arrangement through which investment costs were recovered. European countries changed this financing arrangement between the mid-1990s and 2010, when they had reached very different stages in the development of their power systems. Some had already built most of their hydro and nuclear capacity, while others still had much of that capital-intensive investment ahead of them. This paper asks how that difference in timing relates to the financing conditions under which countries now add new generating capacity.

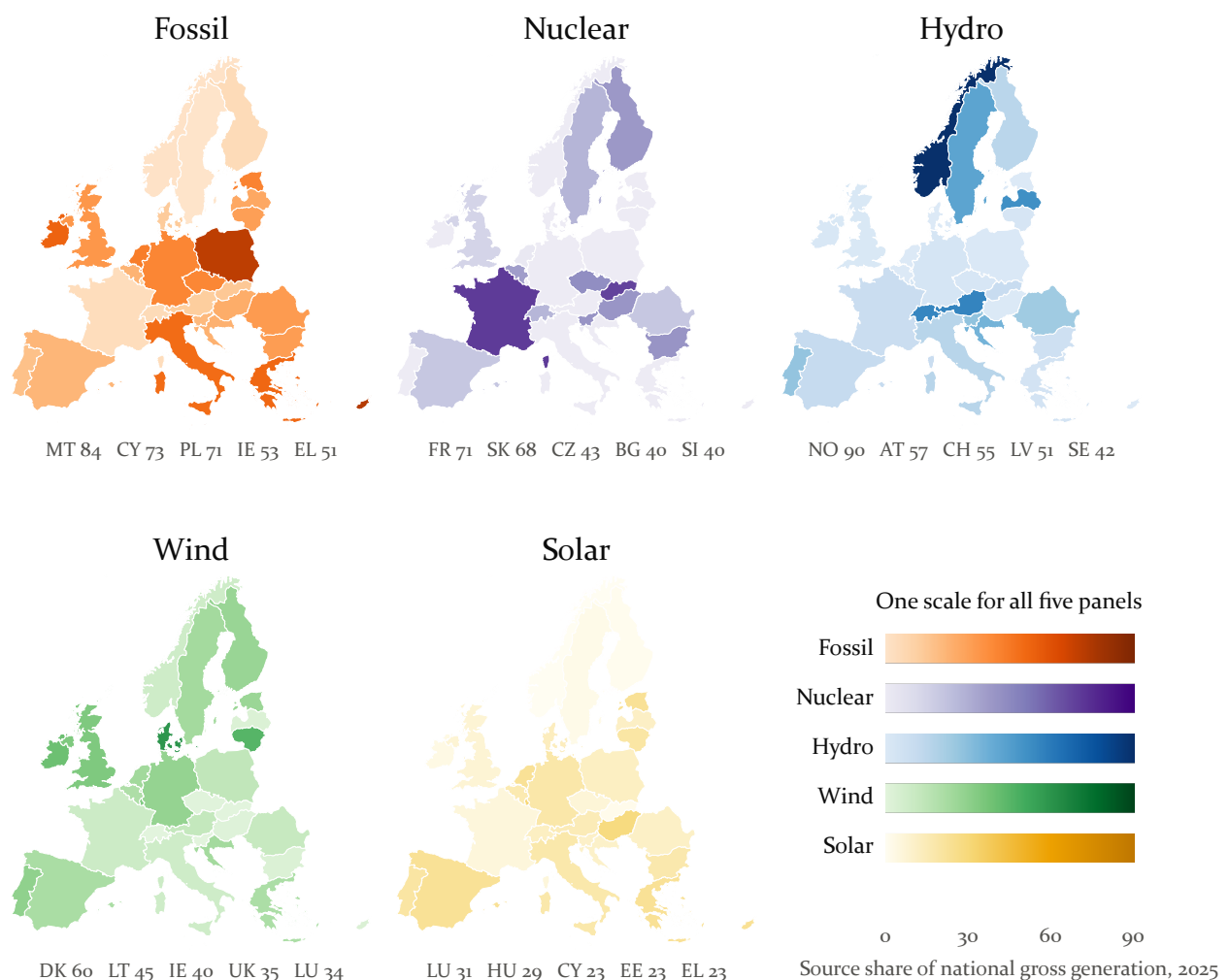


Figure 1. Where each source is a large share of a country's own generation, 2025. One scale for all five panels, so the pale solar panel beside the dark fossil panel is the finding. The five highest countries are named under each; every country and source is in Figure 8.

1. One market, thirty power systems

The single market prices one commodity across thirty physical systems that have little in common. In 2025 the fossil share of generation was about 1 per cent in Sweden and Norway, 70.9 in Poland and 84.4 in Malta. Nuclear supplied 70.6 per cent of French generation and none at all in sixteen of the thirty countries. Hydro supplied 90.2 per cent in Norway and none in five countries, wind 59.7 per cent in Denmark and none in three, and solar 31.1 per cent in Luxembourg (Figure 1; every country and source in Figure 8).

Those generation mixes reflect power systems built from different technologies and at different times, by countries that now invest through the same European electricity market. Section 4 measures the inherited capital stock behind them directly. The reported conditions for adding to that capital also differ widely. The nominal after-tax cost of capital reported for renewable projects runs from 1.93 per cent in Germany to 10.59 in Latvia, with Poland at 5.10, Romania at 7.59 and Hungary at 8.55. Each country's figure pools every solar and onshore wind observation from 2018 onward, which is stated here because the two technologies are financed on different plant risk. Pooling serves here as a country-level summary of the available renewable finance observations rather than a technology-specific one, and it keeps no country resting on a single estimate. Table D1 reports the two technologies separately.

A separate earlier survey shows a similar cross-country pattern. An independent survey of onshore wind financing in 2014 covered twenty-five countries and ran from 4.0 per cent in Germany to 12.8 in Greece, and the observations for 2018 to 2020 cover twenty-four of the same countries and run from 2.0 in Germany to 9.1 in Latvia. The whole distribution moved down by about three points as interest rates fell, and its shape survived. The rank correlation between the two dates is 0.63. The five cheapest countries are the same five at both, Germany, Denmark, France, Belgium and the Netherlands. The most expensive are in the south and east at both, Greece, Croatia, Hungary, Romania and Slovenia in 2014, and the three Baltic states, Romania and Croatia five years later (Angelopoulos et al. 2016; Steffen et al. 2025). The level of reported financing costs fell with the interest-rate environment, and large cross-country differences remained.

Section 6 shows what that difference implies for an otherwise identical plant. Under the financing assumptions used there, the modelled levelised cost of solar electricity is USD 23 per megawatt hour under German conditions and USD 44 under Latvian ones. The same rise in the cost of capital moves solar, wind and nuclear several times as far as it moves gas. The rest of this section asks where in the financing chain that gap appears.

The gap widens as finance approaches the project, and only a small part of it is present in sovereign borrowing. In 2018 and 2019 the German and Latvian ten-year government borrowing rates differed by 0.55 percentage points and their corporate borrowing rates by 1.56. The difference widened to 2.85 points on the project loan and to 8.66 points on the project's overall cost of capital (Table D1). Across the fifteen countries observed at every rung, government borrowing rates span 3.31 percentage points and the rates banks charge companies on new loans 2.79, against 8.66 for the cost of financing a project. Those cross-country spans put the project rung 3.1 times wider than the corporate rung and 2.6 times wider than the sovereign rung (Figure 2). Government borrowing conditions therefore capture only part of the cross-country difference, and the dispersion is much larger in the project estimates.

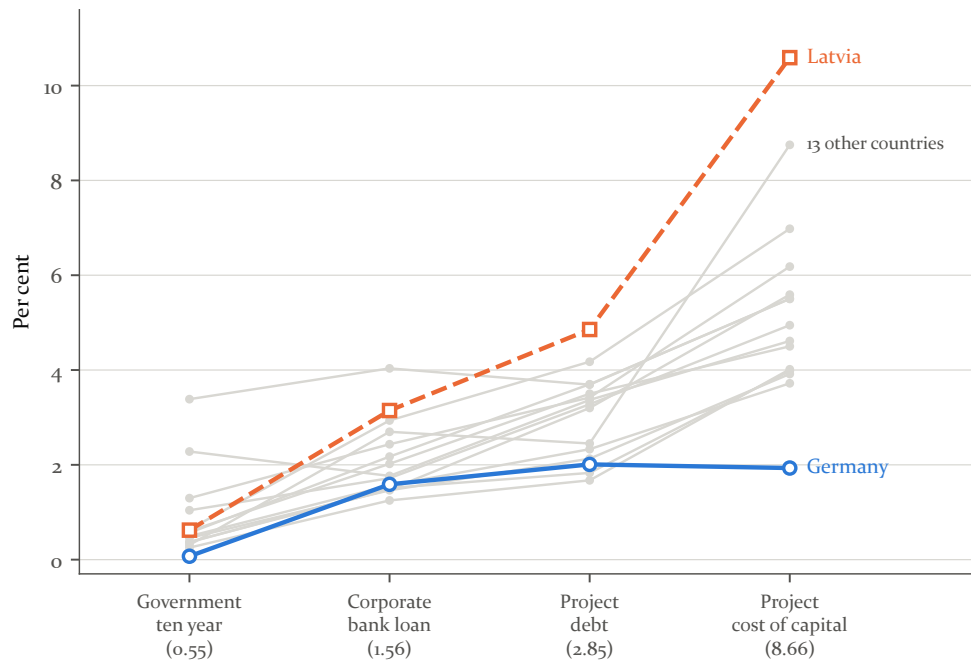


Figure 2. The financing gap widens at every rung. Four measures of the price of money in the same country, for the fifteen countries observed at all four. The first two rungs are ECB series averaged over 2018 and 2019 and are prices at which transactions took place; the project rungs pool every solar and onshore wind estimate from 2018 onward and are survey figures, and the two kinds of evidence are kept apart for that reason. Brackets give the Latvian rate less the German one. Table D1 gives the two technologies separately.

Equity is central to that widening, through both the share of the project it finances and the return it must promise. German projects were reported as financing about 82 per cent of their capital with debt, leaving 18 per cent as equity, against 55 per cent debt and 45 per cent equity in Latvia. Investors in Latvia required a reported return of 19.53 per cent against 4.82 in Germany. The Latvian estimates therefore carry about two and a half times the equity share of the German ones, together with a required equity return about four times as high. The project loan rate moved less, at 4.86 per cent in Latvia against 2.01 in Germany. These component figures are expert survey estimates, averaged separately from the blended cost of capital, so combining them reproduces the blend only approximately (Table D1).

Across Europe the countries with larger established wind and solar markets also tend to have lower required returns on project equity, although the evidence does not establish which causes which. Europe has integrated the market in which electricity is sold, while the available evidence continues to show large differences in the financing conditions for new generating assets. The rest of the paper sets those differences against the history of how national power systems were financed and when their financing regimes changed.

2. How the old system financed investment

Before liberalisation, electricity in most European countries was supplied by vertically integrated utilities operating within protected monopoly territories. Ownership varied, from national and municipal utilities to privately owned companies, while entry and consumer choice were tightly restricted (Newbery 2005). All twenty-three European countries with a long OECD regulation series scored between 3.84 and 6.00 in 1990, and twenty scored exactly 6.00, the value for a closed monopoly with no third-party entry (Table C1). That index describes the organisation of the market, meaning entry barriers, vertical integration, market structure and public ownership, rather than the size or ownership of the utility that built an individual power station.

Those countries did not finance their power systems in the same way. Capital came from state budgets, public borrowing, utility balance sheets, consumers and industrial shareholders. What the arrangements shared was a route to repayment rather than an owner or a source of capital. The utility, or in Finland its contracted shareholders, could recover the cost of investment from electricity users without depending on what an individual plant might earn in a future wholesale market. The regulatory literature calls this relationship the regulatory compact,

under which a utility accepts an obligation to serve and price regulation in exchange for a protected franchise and an opportunity to recover its prudently incurred costs (Joskow 2008).

France shows how the source of capital could change once the utility had an established revenue base. Prices were set at the development cost of a set of plants plus a return on capital of 8 to 9 per cent, so consumers carried most of the risk and EDF could expect to recover what it spent. The hydro programme of the 1950s and 1960s was financed principally through public credit, 85 per cent of it through the Fonds de développement économique et social. The nuclear programme was financed differently. By the early 1970s EDF was meeting 63 per cent of its investment from its own resources, and the state's capital endowment in 1970 was 550 million francs against more than 5.1 billion of investment. By 1973 the Treasury contributed nothing at all. EDF raised the two billion francs not covered by self-financing in two market loans, taken without recourse to the state guarantee (SFEN 2022, citing Bergougnoux 1999).

From 1974 the programme moved from retained earnings to borrowing, so that over 1955 to 1985 as a whole borrowing covered the larger part of EDF's financing need. The Messmer plan launched thirteen reactors at once, and retained earnings could not carry an expansion of that size. The loans carried no state guarantee and no contractual obligation on the French state. EDF raised them on a triple-A rating, at a real interest rate of about 1 per cent, although the source notes that the rating owed something to investors' sense that the government stood behind the company.

The French state remained central to the electricity system in both phases. EDF remained wholly state-owned, and the tariff that made self-financing possible was itself a regulatory grant. What changed was the source of the marginal capital. The first fleet was financed with public credit, and the second could increasingly be financed against the revenues and the balance sheet of the utility itself. France therefore ran one financing regime for its hydro fleet and another for its nuclear one, and it is assigned in Table C2 to the regime that governed its principal capital-intensive build, the 1980s, when 61 per cent of that capacity was commissioned.

The other cases show that there was no single European model. Norway put much of the capital through the national budget and supplemented it with domestic, foreign and World Bank borrowing. The state utility took up no borrowing of its own while it remained a state directorate, which it did until 1985. The Treasury therefore raised the earmarked World Bank loans on four hydro projects and re-lent them through the ordinary appropriation process (Jensen and Vognild 2023). Generation and transmission investment was subject to cost reimbursement, and Parliament approved the state utility's price in the annual budget, a price that reflected average generation cost until 1979 and long-run marginal cost thereafter (NVE and Norad 2017). Britain nationalised the industry in 1947 and financed it through public borrowing capped each year by an External Financial Limit. In some years that limit was negative, so the industry returned funds to the Treasury. The Central Electricity Generating Board sold its output to twelve area boards through a Bulk Supply Tariff (Newbery 2005).

German utilities operated behind protected area monopolies. The Energy Industry Act of 1935 governed the sector until 1998, large vertically integrated utilities owned central generation and networks alongside municipal ones, and customers could not choose their supplier and paid tariffs approved by the state authorities (Agora Energiewende 2019). The evidence establishes the revenue arrangement more clearly than the capital route. RWE had municipal and private shareholders and VEBA was a state-owned holding, and no federal fund or guarantee for their investment existed, which is consistent with financing through their own earnings and their own borrowing. I have not found a document that states the capital route, and Table C2 marks the German cell partly documented for that reason.

Denmark and Finland raised the capital outside the utility altogether. Denmark's sector was owned from below, by more than two hundred distribution companies, each a consumer cooperative or a municipal company, which owned the generating companies between them. The 1977 Electricity Act required the sector to operate on a non-profit basis and allowed a company to collect 75 per cent of the cost of new capacity from consumers before it was built. Denmark therefore built its fleet out of prepayments rather than debt. By 1995 its electricity companies were valued at 15 to 20 billion dollars against roughly one billion of debt (Danish Energy Agency 2020). Finland built its large plants through the Mankala model, a company that pays no dividend and sells its output to its own shareholders at cost, each shareholder entitled to a share of the electricity and obliged to pay the matching share of the costs including the loans. The plant therefore has a contracted offtaker for its whole output before it is

built, the Supreme Administrative Court confirmed the arrangement in two decisions in the 1960s, and almost every large Finnish plant since, the Olkiluoto reactors among them, has been financed this way (TVO 2023).

Country	Who owned the utility	Where the capital came from	How it was repaid
France	EDF, in state ownership	About 63 per cent from EDF's own resources, the rest borrowed on EDF's own credit without a state guarantee; state capital under a tenth in 1970	A tariff EDF was free to set at a level that covered its costs
Norway	A state directorate until 1985, beside municipal and private companies	The national budget, contributions from municipal owners, domestic and foreign commercial loans, and World Bank co-financing on four hydro projects	Cost reimbursement, at a price Parliament approved in the annual budget
United Kingdom	State ownership from 1947	Public borrowing, capped each year by the External Financial Limit, which was negative in some years	The Bulk Supply Tariff, paid by twelve area boards
Germany	Mixed, large integrated utilities beside municipal ones	The utilities' own balance sheets, behind an area monopoly	Tariffs approved by the state authorities, with no supplier choice
Denmark	Consumer cooperatives and municipalities, owning the generators between them	Consumers in advance, since a company could collect 75 per cent of the cost of new capacity before it was built	Non-profit under the 1977 Electricity Act, with only necessary costs in the price
Finland	A jointly owned company paying no dividend, the Mankala model	Shareholder equity and the company's own borrowing, each shareholder liable in proportion to its holding	Shareholders take the output at cost and are obliged to pay the matching costs

Sources as in the paragraphs above: Newbery (2005) for the United Kingdom, Jensen and Vognild (2023) and NVE and Norad (2017) for Norway, Agora Energiewende (2019) for Germany, Danish Energy Agency (2020) for Denmark and TVO (2023) for Finland. These are the six countries the sources reach at this level of detail. Table C2 places all thirty in five financing families at a coarser grain, and says of each cell whether it carries a document or the general account of its group.

The six cases differ in where investment capital came from and agree in how its cost was recovered. EDF's own resources, the Norwegian state budget, British public borrowing, German company balance sheets, Danish consumer prepayments and Finnish shareholder contributions are six sources of money. In each case the arrangement that serviced the money provided a route to cost recovery that did not depend on what an individual plant might earn in a future wholesale market.

Finland separates ownership from cost recovery most clearly. It reached the same result through a contract among shareholders rather than a regulated tariff, with no state ownership and no regulator setting the price. Ownership alone is therefore insufficient to classify these arrangements. France was state-owned and Germany mixed, Denmark cooperative and municipal, Finland a shareholder consortium, and what mattered for financing was whether the capital provider faced the revenue risk of a single plant selling into a future market.

The first fleet could finance the second

Financing regimes were not permanent national characteristics. In three documented cases they changed as the electricity system accumulated assets and cash flow.

Country	First fleet	Second fleet	What changed
France	Hydro, 1950s and 60s. Family B: 85 per cent financed by the FDES, the state's development fund (SFEN 2022)	Nuclear, 1970s and 80s. Family A: 63 per cent from EDF's own resources, the Treasury withdrawn entirely by 1973, the balance borrowed on EDF's credit without a state guarantee (SFEN 2022, citing Bergougnoux 1999)	Public credit to the utility's own balance sheet, once the first fleet was generating cash
Finland	Hydro, 1950s. Family B: the Bank lent to the Mortgage Bank of Finland under a Finnish Government guarantee, which relented to the companies, with state undertakings to cover any balance (IBRD 1956)	Nuclear and large thermal from the 1970s. Family D: the Mankala company, which pays no dividend and sells its output to its own shareholders at cost (TVO 2023)	State-guaranteed credit to an at-cost consortium of the plant's own offtakers
Hungary	Thermal and Paks nuclear, to the 1980s. Family E: Paks is recorded as the only state investment in the subsector in ten years (World Bank 1986)	The trust's other investment by 1986. Sixty per cent of depreciation retained, a Development Fund formed from post-tax profits, and targets set to reduce dependence on the state budget (World Bank 1986)	Plan allocation for the flagship plant, enterprise self-financing around it

France and Finland show the mechanism most clearly. Each built its first capital-intensive fleet in the 1950s on public or publicly guaranteed credit. Roughly two decades later, those systems held operating assets, customers and cash flows against which another generation of investment could be financed, and the second fleet was financed through the utility or its customers rather than the state. Hungary moved in the same direction inside a planned economy, with the flagship plant allocated through the plan and the rest of the trust's investment increasingly self-financed by the time the Bank appraised it.

In these cases an earlier fleet, once operating and generating revenue, could also serve as a financing base for the next one, through the utility or through its customers. Three cases are too few to establish this as a general European sequence. They identify a mechanism that can be compared with the cross-country pattern, which is what Section 4 does. Croatia and Slovenia may be a fourth case, since the self-management reforms after 1971 are said to have moved Yugoslav enterprises towards internal funds and earmarked charges, but that is not sourced here and is left out.

The institutional sequence after 1990 was different in the former planned economies. The World Bank's 1998 sector strategy for Europe and Central Asia is the document under which it lent to the region through the transition. It lists low cost recovery among the six defining problems of the power industry. It then commits the Bank's lending to competition in generation and supply, to market-based pricing with minimal regulatory intervention, and to a shift of government "from that of owner/manager to policymaker, at the same time that the government sells assets in the power industry to private, strategic investors", with Bank guarantees against non-commercial risk to help the sales along (World Bank 1998). The documented sequence was therefore different from the one observed in France and Finland. These economies moved from plan allocation towards market-based investment without the intermediate period in which an earlier publicly financed fleet supported later utility or customer financing. Whether that difference left them with more investment to finance after liberalisation cannot be established from these historical cases, and Section 4 examines the proposition using the capital stock data.

The distinction between who supplies the capital and who bears the revenue risk is central to what follows. A state can own a utility without financing its investment from the government budget, as France shows, and private shareholders can own a utility whose investment is supported by regulated cost recovery, as Germany shows. Regulated cost recovery reduced the exposure of investment revenue to wholesale electricity prices and allowed an established utility to use the cash flow of existing assets to help finance new ones. Liberalisation changed the revenue stream against which new investment had to be financed.

3. What liberalisation changed, and when

The single market directives dismantled the system of vertically integrated monopoly and regulated cost recovery described above. The European mean of the OECD regulation index fell 4.28 points from its 1990 level. Of that decline, 61.5 per cent occurred between 1995 and 2005 and 93.5 per cent was complete by 2010. Since 2010 the index has moved another 0.28 points in thirteen years (Table C1). Most of Europe's electricity market liberalisation was therefore concentrated in the fifteen years between the mid-1990s and 2010.

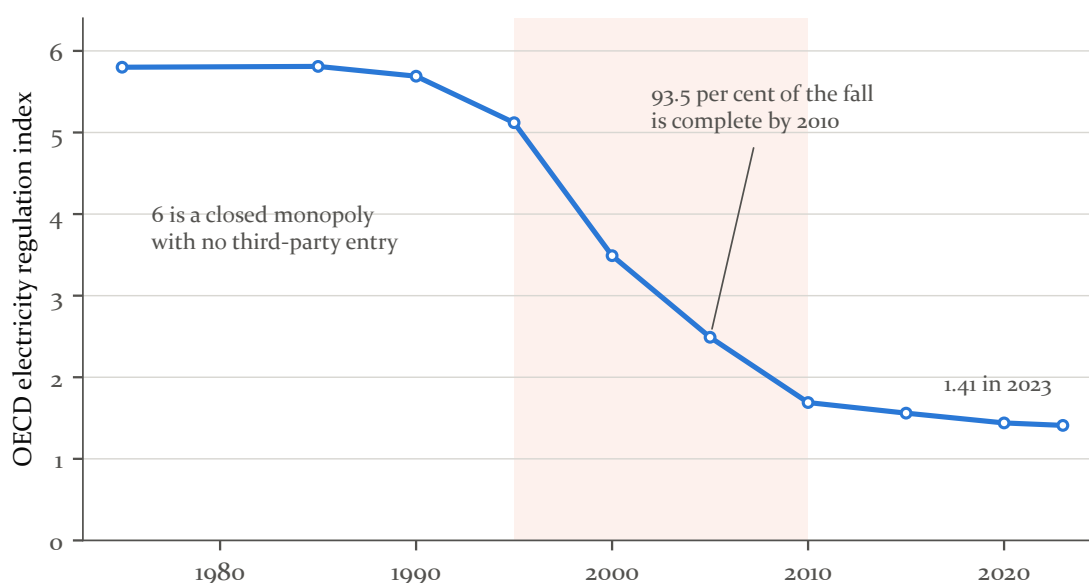


Figure 3. When the financing regime changed. The mean OECD electricity regulation index across the twenty-three European countries with a series back to 1975, on a scale where six is a closed state or municipal monopoly with no third-party entry. The shaded band is the period this section describes. Table C1 gives the index country by country and each country's own reform year.

Liberalisation increased generators' exposure to market revenues as vertically integrated regulated systems were unbundled. Existing research finds that wholesale electricity revenues alone provide too little certainty to finance some long-lived capital-intensive assets. Energy-only markets may fail to recover fixed costs, and the long-term contracts needed to support investment do not necessarily emerge on their own (Newbery 2016). Lenders and shareholders therefore seek longer revenue certainty than the short-term market provides. European electricity markets have developed further arrangements alongside the short-term market. Power purchase agreements and contracts for difference give investors greater certainty over future revenue, capacity mechanisms pay for availability where security of supply requires it, and grants, public loans and European funding programmes reduce the amount or the price of the capital a project has to raise. These instruments perform different functions, and each addresses a financing need that the wholesale price alone does not meet (Keppler, Quemin and Saguan 2022; Roques and Finon 2017).

Member states can still support the construction of new power stations, and liberalisation changed how that support is provided. Unbundling separated regulated electricity networks from competitive generation, so the construction cost of a new generator could no longer be recovered through the regulated revenues of an integrated utility (Directive (EU) 2019/944, Articles 35 and 43). Support for generation instead requires a separate mechanism built around the project. Hinkley Point C illustrates the new model, where the contract for difference, the Secretary of State agreement and the credit guarantee required a European Commission state aid decision in

2014, upheld by the Court of Justice in 2020 (Commission Decision (EU) 2015/658). Revenue protection therefore survived liberalisation in a different institutional form. Under the arrangements of Section 2 it was embedded in the utility, its customers or its contracted offtakers. Under the liberalised system it is generally attached to an individual investment or class of investment.

The instruments that do this act on different quantities. A grant reduces the amount of capital a project has to raise. A public or concessional loan reduces the price of its debt. A contract for difference reduces the uncertainty attached to its future revenue. Regulated cost recovery worked on a different level again, because it applied to a utility's whole investment programme and to its customer base rather than to one project at a time. EDF did not negotiate a separate thirty-five-year contract for each reactor, and it could put the cash flow of depreciated hydro plant behind the next one. A grant lowers what a project must raise and a contract for difference protects what it earns. Neither reproduces the balance sheet of an established regulated utility. Liberalisation separated functions that had previously been combined. Grants, public loans and contracts for difference act on different parts of the resulting financing problem. How those institutional differences relate to the financing conditions reported in Section 1 is what the rest of the paper examines.

The argument of this paper concerns when that change occurred relative to what each country had already built. European countries liberalised over broadly the same period, and they entered that period with very different inherited power systems. France had already built most of its nuclear fleet under regulated cost recovery, and Norway most of its hydro fleet. Poland remained heavily dependent on coal, while the Baltic states inherited electricity systems built under Soviet institutions. France and Norway therefore financed much of the capital-intensive generation they use today before liberalisation. Countries now replacing coal and Soviet-era assets are undertaking a larger share of their capital-intensive investment under the regime that followed. Those countries are not unsupported. They have access to European grants, concessional lending and risk-sharing instruments that France and Norway did not have when they built. Whether those instruments reach the terms on which capital is actually offered is a separate question, and the evidence in Section 1 bears on it.

The French and Norwegian cases also show that the historical financing route cannot be read from government debt or national wealth. France launched the Messmer plan in 1974 with central government debt at 14.5 per cent of GDP, and financed the nuclear programme principally through EDF's cash flow and balance sheet rather than through the government budget. Norway developed its hydro fleet out of a domestic credit market in which borrowing was rationed by annual quota, with a state utility that could not borrow in its own name and with World Bank capital on several projects that would otherwise have been unavailable (IMF Historical Public Debt Database; Jensen and Vognild 2023). What France and Norway possessed was a mechanism for recovering large electricity investments from consumers, together with decades in which to use it before liberalisation changed how new generation was financed.

Liberalisation was also a different thing from privatisation. Between 2018 and 2023 the state's holding in the largest generator increased in France, Poland, Ireland and Finland, fell only in Greece, and was unchanged in the other eighteen countries (Table B2). Sweden and Norway make the distinction particularly clearly, since their OECD regulation indices fell by 4.75 points while their largest generators remained wholly state-owned. A government can therefore go on owning the generator after liberalisation, while the market and revenue structure within which that generator invests has changed.

4. What was already built when the regime changed

The capital stock measure separates what a country had already financed when the regime changed from what it accumulated afterwards. Eurostat reports the stock of fixed assets held by NACE sector D, electricity, gas, steam and air conditioning supply, as chain linked volumes in 2020 euro, which makes a country comparable with itself across years. Divided by the electricity the sector produced, it gives the capital in place per unit of output. Twenty-two European countries report that series for 2000, 2010 and 2023.

The two periods divide sharply. Between 2000 and 2010 the growth of a country's energy capital stock was unrelated to how much capital it already had, at a correlation of -0.07 between annual growth and the log of the starting level. Between 2010 and 2023 the correlation is -0.53 , on a slope of 1.64 percentage points of annual growth lost for each log point of starting stock. Countries entering the second period with less capital per unit of electricity built faster in it, and no such pattern is present in the decade before.

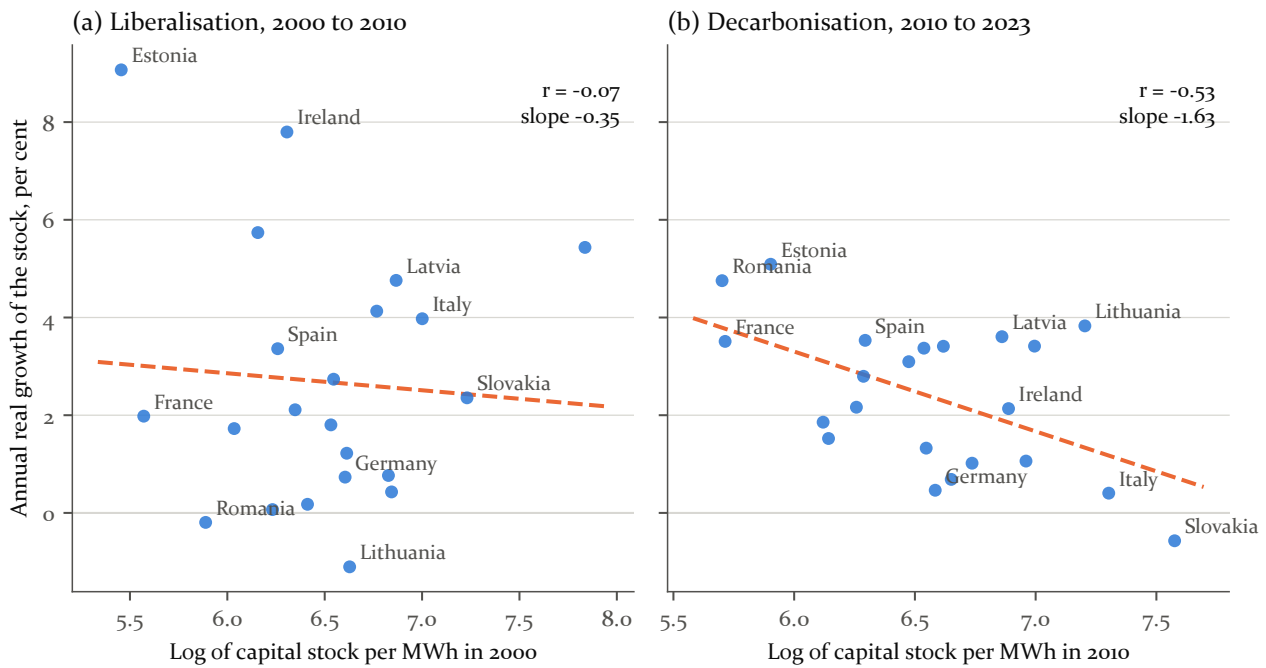


Figure 4. Where capital formation went, before and after. Annual real growth of the sector D capital stock against the log of the stock per megawatt hour at the start of each period, for the twenty-two countries with a chain linked volume series. Each panel gives the correlation and the fitted slope. The fitted line is flat in the first period and clearly downward in the second, and that difference is the claim. Ten countries are named, the rest are plotted unlabelled. Table A2 holds the series for every country, the robustness checks and the coverage notes.

A mechanical small-base effect is not consistent with the change, since it would operate in both periods and the relationship is absent before 2010. The same holds for regression to the mean and for measurement error in the starting level. The result also survives conditioning on the level ten years before the window opens, entering system size alongside the starting level, and dropping any single country, which holds the correlation between -0.42 and -0.64 . Table A2 reports these checks. France and Spain are the largest positive outliers after 2010, at 3.5 per cent a year each, the first on the refurbishment programme for the existing reactor fleet and the second on renewable additions at a scale no other country in the sample matched.

The result is a change in the geography of capital formation after 2010. Countries doing more of their building since then are those that entered the period with less capital already in place, and they are accumulating more of their capital under the post-liberalisation regime of Section 3. Estonia at 5.1 per cent a year, Romania at 4.8, Lithuania at 3.8 and Latvia at 3.6 lead that group; Slovakia, Italy, Germany, Hungary and Austria are the slowest.

How that pattern relates to financing cost is much less precisely estimated. Across the thirteen countries with both a capital stock series and a reported project cost of capital, the two are positively related at 0.33, which is weak in a sample of that size and does not support a general statement that the least built pay the most. The named cases are sharper. Estonia and Latvia are two of the four fastest builders since 2010 and are reported at 7.64 and 10.59 per cent nominal after-tax cost of capital on a renewable project, against 1.93 in Germany, whose stock grew at 0.46 per cent a year, third slowest of the thirteen. Rapid capital accumulation can therefore coincide with high reported project financing costs.

One measurement point belongs with the figure. A gross capital stock is the output of a perpetual inventory model whose assumed service lives each country sets for itself, so a higher level can mean a longer assumed life rather than more plant. The figure therefore plots growth rates and not levels, since a service life a country holds constant cancels in a within-country growth rate.

5. When inherited capital does not carry forward

Lithuania shows why inherited generating capacity cannot be treated as a fixed endowment. Its capital-intensive legacy was the Ignalina nuclear plant, whose two RBMK units closed at the end of 2004 and at the end of 2009

as a condition of accession, on reactor-safety grounds rather than as part of market reform. Over the same period Lithuania unbundled the state utility that had held the plant and restructured the sector under the European framework. Lithuania can still support a new plant, and any support now has to be cleared by the Commission and priced against the market. Lithuania therefore lost a large generating asset during the same period in which its sector financing arrangements were changing, and it now imports 63 per cent of the electricity it consumes.

Latvia shows a second limitation of inherited capital, which is scale. On the face of it Latvia is in the position France was in when the Messmer plan began: it owns the Daugava hydro cascade, the plants are state-owned and long since paid for, and they supplied 51 per cent of Latvian generation in 2025. The difference is scale. The cascade is about 1.5 GW in a system that generates 5.8 terawatt hours a year, so it is small relative to the investment a new fleet requires, and Latvia is reported with the highest cost of capital in the sample. Size also limits what can be built.

One gigawatt of nuclear running at an 80 per cent load factor would produce more electricity than Latvia or Estonia consumes in a year. A written-off fleet also has a first claim on its own earnings, since it has to be kept running, and France's 3.5 per cent a year since 2010 is the refurbishment of the existing reactors rather than new plant. An existing fleet is therefore a financing base only to the extent that its revenue, after operation, maintenance and renewal, leaves capacity for additional investment, and only when it is large relative to what comes next.

6. What financing does to the cost of power

Capital-intensive technologies are much more sensitive to financing costs than fuel-intensive ones. Solar, wind and nuclear pay most of their costs upfront, while gas pays much of its cost later through fuel. The cost of capital is the return required by the lenders and shareholders who provide that upfront money. The levelised cost of electricity converts the plant's investment, financing, fuel and operating costs into a cost per megawatt hour over its life.

Figure 5 shows the size of the effect on one technology. One benchmark solar plant, IRENA's global average installed cost and capacity factor, is priced at each country's pooled cost of capital from Figure 2, and nothing else is allowed to vary, which is the design Egli, Steffen and Schmidt (2018) used. The modelled levelised cost of the same plant is USD 23 per megawatt hour under German financing assumptions and USD 44 under Latvian ones, a ratio of 2.0. Three qualifications belong with it. Each figure is the average revenue per megawatt hour needed over the plant's lifetime to recover the modelled costs, including the assumed return on capital. None is an observed sale price, a contract price or a consumer price, all of which are set by the wholesale market, network charges, taxes and supply contracts. It follows that the comparison says nothing about realised profits, since no comparable record exists of the prices or contracts Latvian and German solar projects actually obtain. And no country's actual solar cost is claimed: real plant costs and solar resources differ by country, and the figure removes that difference deliberately so that the whole of the bar is financing.

Observed German utility-scale costs are higher than the 22 shown here, because German irradiation is below the global average the benchmark uses; the ratio is unaffected, since the cost of capital enters the same way whatever the plant produces. The Latvian rate is the mean of four survey estimates across the two technologies, used as an illustrative high financing cost rather than a precise national average.

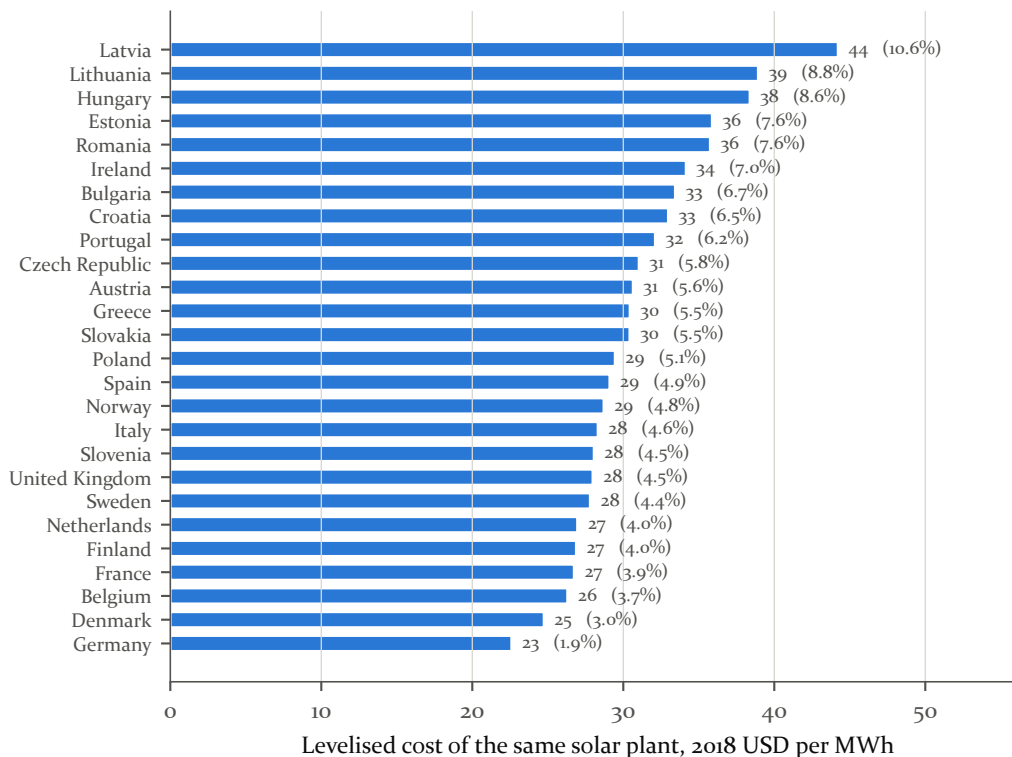


Figure 5. The same solar plant priced at each country's reported cost of capital. The plant is IRENA's global-average benchmark and the solar resource is held constant, so the bar isolates financing and makes no claim about any country's actual solar cost. The figure in brackets is the nominal after-tax cost of capital for financing years 2018 onward, converted to real at 2 per cent inflation before the plant is priced. Modelled levelised costs in 2018 USD. Sources and assumptions in Tables D1 and D2.

The same arithmetic runs through time, and it can disadvantage an earlier project when equipment costs later fall. A plant financed in Latvia in 2019 carries its 10.6 per cent for the whole of its life, because the capital charge is fixed at financial close, while a later project may face lower equipment costs and different financing conditions. In a market that pays every plant the same hourly price, both receive that price, and it can fall in the hours in which solar runs as more solar enters. The price the earlier plant receives can therefore fall while the capital charge fixed at its financial close does not. Only a contract fixed at financial close, a feed-in tariff or a contract for difference, holds that charge against the market price, which is what Germany's early solar and wind had under the EEG. Long-term revenue arrangements therefore matter alongside financing cost, because they decide how much market-price risk remains with a project after financial close.

Figure 6 shows that the same increase in the cost of capital moves technologies by different amounts. Raising the cost of capital from 3 to 9 per cent increases the modelled cost of solar by 57 per cent, onshore wind by 37 per cent and nuclear by 87 per cent, against 19 per cent for coal and 10 per cent for gas. Under these assumptions the modelled nuclear cost crosses the modelled gas cost at a real cost of capital of about 9 per cent.

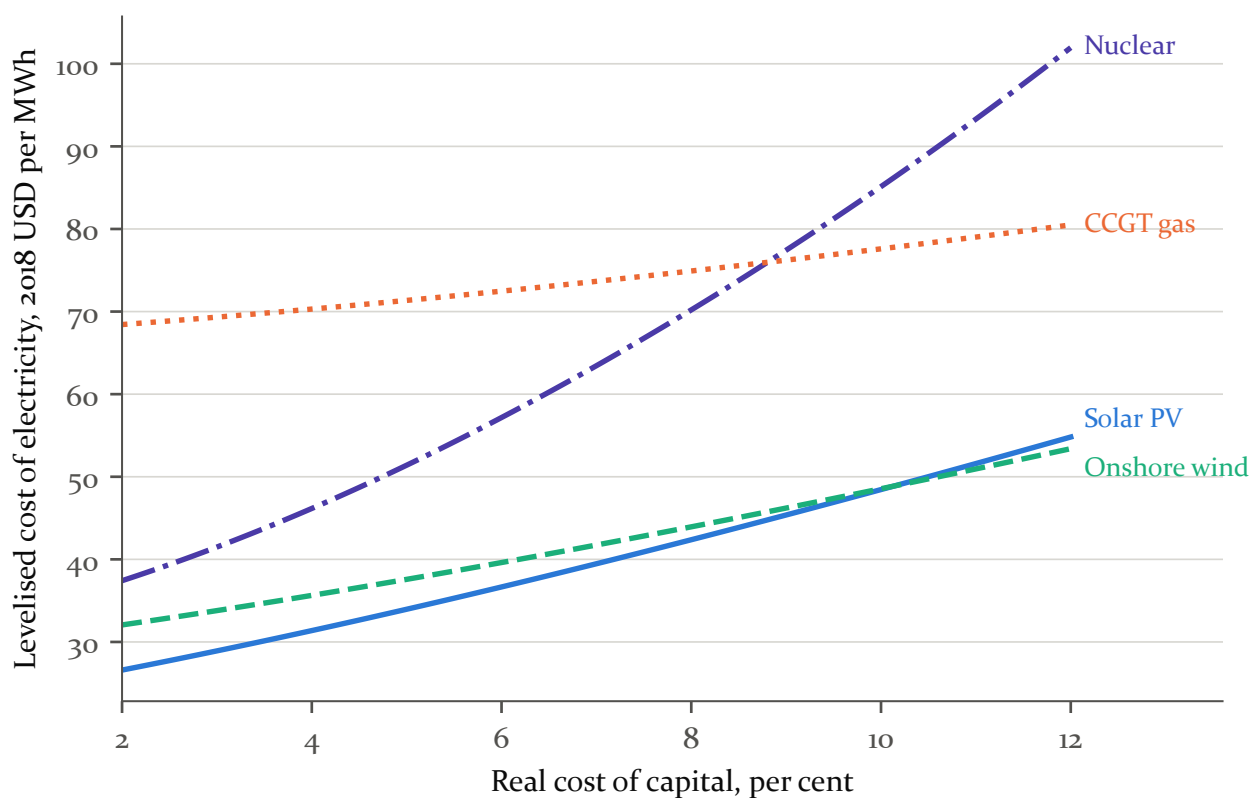


Figure 6. Financing reorders the technologies. A sensitivity calculation, not an estimate of observed generation costs: the plant, its fuel and its carbon charge are held fixed and only the cost of capital moves. The point at which nuclear crosses gas is an output of the model rather than a published result, and it moves with the construction cost assumed. Every input, the constructed benchmark it rests on and the check on the model are in Table D2.

Figure 7 shows why. At a 6 per cent cost of capital, recovery of the upfront investment accounts for 81 per cent of solar's levelised cost, 59 per cent of nuclear and 55 per cent of onshore wind, against 21 per cent for coal and 14 per cent for gas. Coal is the useful intermediate case, because it carries both a large construction cost and a large fuel and carbon bill, which is what shows that the mechanism is about when costs fall due rather than about fossil fuels against renewables.

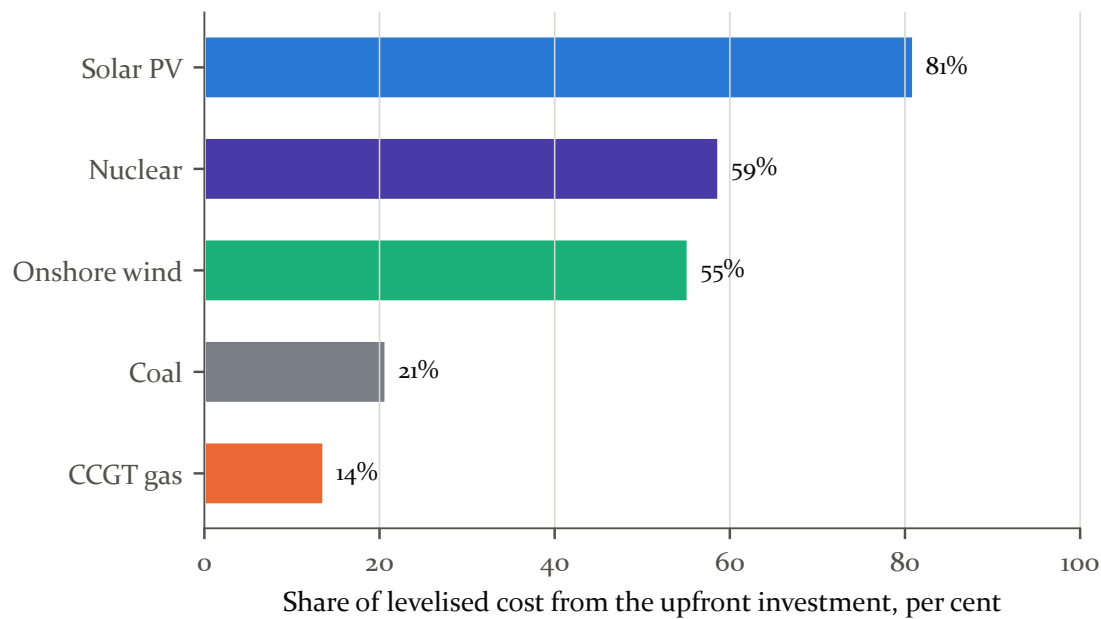


Figure 7. Technologies differ sharply in how far they depend on upfront investment. The share of levelised cost accounted for by recovery of the upfront investment, at a 6 per cent real cost of capital. Coal appears here and not in Figure 6 because it carries both a large construction cost and a large fuel and carbon bill, and is therefore the case that separates when costs fall due from whether a technology burns fuel. Table D2.

Both figures are sensitivity calculations rather than estimates of observed generation costs. The plant, its fuel and its carbon charge are held fixed and only the cost of capital changes. The technology cases combine published benchmark assumptions from IRENA for solar and onshore wind with IEA/NEA assumptions for nuclear, coal and gas, expressed in common 2018 US dollars. They are a constructed benchmark rather than a dataset published by either organisation, and the point at which nuclear passes gas is an output of the model rather than a finding of the IEA/NEA study. Table D2 gives every input with its unit and its source, and reports how closely the model here follows the IEA/NEA method.

These sensitivity calculations matter because the largest cross-country dispersion in the available financing evidence appears at the project level. Across the fifteen countries observed at every rung, the range widens from 3.31 percentage points on government borrowing and 2.79 on ordinary corporate bank borrowing to 3.18 on project debt and 8.66 on the project's overall cost of capital (Figure 2; Table D1).

That widening occurs through more than the project loan rate. Project finance combines debt and equity, and equity normally requires a higher return because it absorbs losses before lenders do, so greater reliance on equity can raise the overall cost of capital much more than the loan rate difference alone. Across the seventeen countries measured against Germany, the required equity return accounts for 43 per cent of the reconstructed gap, the project loan rate 35 per cent and gearing 21 per cent (Table D1). The observations behind that decomposition remain thin, so it applies to the available estimates rather than to representative national financing structures.

The renewable finance literature makes the same distinction. It separates general financing conditions from country, policy, technology and project specific risks (Steffen 2020). Country risk in that literature is broader than the government's borrowing rate, because investors also weigh the regulatory framework, the credibility of the revenue arrangement, experience in the local market and the other risks that decide whether and on what terms a project can be financed at all. Those risks appear in the project loan rate, in the amount and maturity of the debt, and in the return equity requires.

Experience and market development matter as well as risk. The survey work behind the project estimates used here finds lower financing costs where a country has accumulated more experience with renewable deployment and with auctions, and it emphasises the terms of project debt, including loan maturity and debt capacity, rather than the interest rate alone (Roth et al. 2021).

Regulated networks provide a comparison, because capital-intensive investment there is financed against a different revenue arrangement. They are financed in the same national capital markets and face the same sovereign. A network operator invests against a regulated asset base and recovers efficiently incurred costs and an approved return from network users over time, so its investors carry much less wholesale market revenue risk than the investors in an uncontracted generator. A contract for difference can give a generator some of the same certainty, through a contract attached to the project rather than through a regulated asset base. The difference lies in where the risk goes. Network regulation transfers much of the revenue risk to consumers, while an uncontracted generator leaves it with investors, and a contract for difference transfers part of it back. Public or private ownership does not determine that allocation (Table B2).

Financing cost is one part of the problem. The transition also raises the amount of capital that has to be committed before generation begins.

Lower generation cost does not mean lower financing need. Solar and wind avoid the continuing fuel costs of gas, which can give them a lower levelised cost of electricity, and they still require more capital to be committed before any electricity is produced. Using the published benchmark investment costs and capacity factors, producing a given amount of annual output requires about 1.9 times as much upfront capital with coal as with a gas turbine, 2.2 times with onshore wind, 2.8 times with solar and 3.5 times with nuclear, rising to 4.2 times for the French reactor that is the IEA/NEA study's European nuclear case (Table D2). These ratios measure how much upfront investment has to be financed for a given amount of annual output. They are not levelised costs of electricity, and unlike a levelised cost they do not depend on how long the plant then runs. The transition from gas to renewables can therefore lower the cost of electricity while increasing the amount of investment that has to be financed in advance.

The number of units is rising as well. After fifteen years of flat or falling consumption, electricity demand is growing again in the advanced economies, and the International Energy Agency forecasts an additional 300 terawatt hours of demand in the European Union by 2030, from electric vehicles, heat pumps, cooling, industry and data centres (IEA 2026). Against the 2 608 terawatt hours the twenty-seven member states generated in 2025 (Table A1), that is about a ninth more electricity. More capital for each unit of output and more units together raise the volume of investment that has to be financed.

Capital intensity alone does not produce a high cost of capital. The financing problem arises when large sums have to be committed for long periods against uncertain future revenues. The transition makes that problem more important because it increases both the amount of capital that must be financed for each unit of electricity produced and the number of units. Regulation and long-term contracts matter because they determine who carries the revenue uncertainty, and therefore the terms on which that larger capital requirement can be financed.

7. Who ultimately pays for investment

Most power-sector investment does not appear as direct government capital formation. It accounts for 7.9 per cent of sector investment in France, 6.1 per cent in Poland, 1.4 per cent in Germany, 0.7 per cent in Spain and none in Denmark. Public support can reach investment through several other channels, including state-owned companies, regulated charges, contracts, grants, guarantees and European institutions. Ownership alone therefore does not identify how an electricity investment is financed or how its cost is recovered.

Networks make one of those channels visible. Their investment is recovered from electricity users through regulated charges, and it is taking a growing role in European power-sector investment, since network investment rose from USD 51 billion in 2015 to USD 144 billion in 2026 while generation investment was broadly flat. The resulting charges are already substantial. Network charges and levies together account for 77 per cent of the 2025 household price excluding VAT in Hungary, 75 per cent in Sweden, 72 per cent in Denmark, 71 per cent in Poland, 62 per cent in Germany and 31 per cent in Greece (Table B1). Those shares are not measures of capital cost, since network charges also recover operating costs and levies include taxes and other policy charges, but they show how much of the electricity bill already sits outside the wholesale energy price.

The capacity to absorb those costs also differs across countries. On a purchasing power basis Romania becomes the most expensive country in the sample and Poland rises nine places, while Germany falls to fourth. Financing conditions and the capacity of households to absorb regulated charges therefore vary across the same European

market, and in several countries they run in the same direction, with a higher cost of capital for new generation alongside less household purchasing power across which regulated investment costs can be spread.

8. What the evidence can, and cannot, show

This paper connects two literatures that have largely remained separate. Research on renewable energy finance measures substantial cross-country differences in project financing conditions and studies their consequences for investment (Steffen 2020; Roth et al. 2021; Polzin et al. 2021; Egli, Steffen and Schmidt 2018). Research on electricity liberalisation examines the timing of market reform and its institutional consequences (Erdogdu 2014; Jaxa-Rozen, Wen and Trutnevskyte 2025). This paper links them by measuring the capital stock already in place when the financing regime changed. It measures how much of each country's power-sector capital was in place at that point, documents how that inherited position differs across Europe, and relates it to subsequent capital formation and to reported project financing conditions.

The contribution is to place present financing conditions in the historical sequence through which national power systems were built. Countries entered broadly contemporaneous liberalisation with very different amounts of capital-intensive generation already financed under earlier cost-recovery arrangements, and the sector D capital stock makes that inherited position observable, which allows the subsequent geography of investment to be compared with it.

The contribution has three empirical parts. The measurement is a cross-country series of inherited power-sector capital at the institutional transition, set against capital formation afterwards. The historical work shows that ownership and the source of capital are insufficient to classify how electricity was financed, since the six documented countries drew on entirely different sources under a common mechanism of cost recovery, and that in three cases the regime itself changed as an earlier fleet accumulated assets and cash flow. The integration is to connect that starting position to a literature that studies present project financing conditions without asking how much of today's system each country had already financed under the preceding regime.

The financing ladder of Section 1 provides the contemporary evidence to which this historical comparison is applied. That renewable financing costs differ across Europe is established, and the IEA and others emphasise that country and macroeconomic conditions are among their determinants. What this paper adds is where those present differences sit within a much longer financing history.

Nothing here identifies a causal channel running from liberalisation to today's financing costs, and the paper does not have the observations that would identify one, since comparable project financing terms are not available for the period before the reform. The paper documents the cross-country relationship between them and the historical sequence in which they arose.

The historical comparison helps clarify what this evidence can and cannot establish. France commissioned 49.6 GW in the 1980s, 96 per cent of it capital-intensive, after committing to most of that investment in the 1970s, when the real long-term rate it faced was 0.21 per cent against 4.39 at commissioning. The United Kingdom commissioned 15.9 GW in the 1990s, 4 per cent of it capital-intensive. The contrast is larger than the difference in long-term real rates alone would suggest, and the historical evidence documents different revenue arrangements against which the investment was committed.

9. What changed after the project-finance evidence ends

The financing evidence used in this paper comes largely from 2018 and 2019, near the end of Europe's low-rate decade. Borrowing has since become substantially more expensive and also more similar across countries. Across the twenty countries observed on both measures, the mean sovereign long-term rate rose from 0.87 per cent in 2018 and 2019 to 3.12 per cent in 2025, and the mean composite cost of bank borrowing for companies rose from 2.24 to 4.02. The cross-country range narrowed over the same years, from 3.31 percentage points to 0.98 on sovereign borrowing and from 2.79 to 1.76 on corporate borrowing, and Latvia's corporate premium over Germany fell to 1.06 points.

Whether project finance converged with them is unknown. Section 1 locates most of the cross-country gap after capital reaches the project, and comparable observations on project debt shares, loan terms and required equity

returns do not extend through the tightening cycle after 2022. Convergence in sovereign and corporate rates therefore cannot establish convergence in project finance.

The higher level matters even though the dispersion narrowed. Section 6 shows that a rise in the cost of capital raises the cost of solar, wind and nuclear more than that of fuel-intensive generation, because a larger share of their expenditure falls before production. Europe is also asking the power system to absorb more capital, as grids expand, fossil assets are replaced and electrification and data centres add demand. Europe therefore faces a larger volume of capital-intensive investment and a higher general level of borrowing costs than in 2018 and 2019. One industry estimate reports the same direction, recording policymakers “extending contracted revenue models beyond the networks to the next stage of the value chain, generation”, and the headroom of Europe’s listed utilities falling from 290 billion euro in 2024 to 221 billion in 2025 (Roland Berger 2026).

The institutional environment has changed as well. During the 2022 energy crisis most member states intervened in wholesale or retail prices, and the World Bank reports that the compensation mechanisms “do not always fully offset the financial impact on these companies”. The same report warns that tighter financial conditions, a higher cost of financing and indebted governments “may impact the competitiveness of new low-carbon capital-intensive technologies and delay their deployment at scale” (World Bank 2023). For investment the relevant issue is therefore both the price of capital and the credibility of the future revenue arrangement against which that capital is committed. Some of the exceptional common financing created after the pandemic is also temporary, since the Recovery and Resilience Facility closes for final payments on 31 December 2026, so the capacity of the public counterparty remains part of the financing environment rather than a uniform European condition.

Where Europe’s public instruments enter the financing chain

The instruments in the table cover only part of the financing chain for European power investment. Section 7 shows direct government capital formation running from 7.9 per cent of sector investment in France to 0.7 in Spain and none in Denmark, and the table does not identify who raises the remaining capital. What it identifies is where selected public instruments enter the financing chain, where their own funding originates, and what financing condition each changes for a project. Section 2 separates who supplies the capital from who ultimately repays it, and the same separation applies to Europe’s instruments today. The table gives the scale of each, where the money is raised, and what it changes for a project. The figures are the Commission’s own, for the 2028 to 2034 budget it proposed on 16 July 2025 and for the instruments already running, and the budget lines are proposals until it is adopted.

Instrument	Scale	Of which energy	Period	Where the money is raised	What it changes for a project
Recovery and Resilience Facility	€723bn	€153.8bn	to 2026	EU borrowing on the capital markets; grants serviced through the EU budget, loans repaid by the borrowing state	Grants cut the capital to be raised; loans supply public credit
National escape clause, defence	€650bn	none	2025 to 2029	National borrowing	Relaxes the deficit rule for one category of spending
Security Action for Europe	€150bn	none	to 2030	EU borrowing, single branded EU-Bonds	Lends to member states at the Union's own credit
Modernisation Fund	€57bn	all	to 2030	Sale of emissions trading allowances	Cuts the capital to be raised, in ten lower-income member states
Innovation Fund	€40bn	most	2020 to 2030	Sale of emissions trading allowances	Cuts the capital to be raised, for first-of-a-kind plant
Cohesion mid-term review	€34bn	€1.2bn	2021 to 2027	Existing cohesion allocations	Redirects grants between priorities
Connecting Europe Facility, energy	€29.9bn	all	2028 to 2034	EU budget and own resources	Cuts the capital to be raised, for cross-border grid
Competitiveness Fund, clean transition	€26.2bn	most	2028 to 2034	EU budget and own resources	Cuts the capital to be raised
EIB guarantee on power purchase counterparties	€0.5bn	all	pilot	EIB balance sheet	Substitutes a trusted counterparty for the offtaker's credit

The energy column is the part of each envelope that reaches energy where an official figure exists. For the Recovery and Resilience Facility it is the member states' own allocations to energy measures, 27 per cent of the €577bn they actually programmed, the balance of the €723bn being loans that were never requested. The two defence rows carry no energy and are shown because they establish what the Union's balance sheet can do. The two allowance-funded envelopes assume a carbon price of €75 per tonne. The escape clause figure is the Commission's estimate if every member state raises defence spending to the permitted 1.5 per cent of GDP. Sources: European Commission, the 2028 to 2034 EU budget, the European Competitiveness Fund and the Connecting Europe Facility proposals, the Modernisation and Innovation Fund pages, the SAFE borrowing page and the national escape clause communication; the Commission's REPowerEU four-year review for the cohesion reprogramming; the Affordable Energy Action Plan for the guarantee pilot.

The largest cross-country differences measured in this paper appear in project debt, in leverage and in the required equity return rather than in sovereign borrowing. Most of the instruments in the table act on the amount of capital a project must raise or on the source that supplies it.

The rows are not all the same kind of money, and the differences matter more than the totals. The allowance-funded envelopes recycle carbon revenue, the budget lines spend appropriations, and two instruments raise capital on the Union's own credit and pass its terms on. Almost all of them act on the amount a project must raise, while the cross-country gap measured here opens in the terms on which the remainder is raised. The guarantee in the last row is the one instrument that acts directly on a project-level risk. Grants may also improve financing terms indirectly, by lifting cover ratios or making a marginal project bankable, and that effect is not measured here.

Several of these instruments concentrate on lower-income member states, among them countries the project evidence reports at relatively high financing costs. Whether European energy support rises systematically with the cost of capital a country faces is not tested here, since eligibility is not allocation and allocation is not financing received.

These instruments also draw on different balance sheets and different forms of fiscal capacity. SAFE shows that the Union can borrow centrally and pass its financing conditions to member states, and the escape clause shows that the deficit rule can be relaxed for a named category of investment. The cohesion mid-term review shows that allocations already made can be moved between priorities. It reprogrammed €34 billion, about a tenth of cohesion funding, of which €1.2 billion went to the new energy priority across eleven countries. Energy support is more fragmented: carbon revenue recycled through two funds, appropriations through two more, national schemes beside them, and the rest raised project by project. None of this establishes that defence spending crowds out energy investment. It does show that the Union and its member states currently use common borrowing, guarantees and fiscal flexibility for specified purposes, and that a guarantee can support a larger nominal volume of investment than an equivalent budgetary provision while leaving the public sector exposed to contingent losses. The narrower question this paper raises is whether comparable risk-bearing capacity reaches electricity investment at the project level, where the cross-country gap is largest.

What Europe does not measure

The main empirical gap is at the project level. Europe observes sovereign yields continuously and corporate borrowing rates monthly, and the evidence assembled here contains no recurring comparable measure of project loan margins and maturities, leverage or required equity returns across member states. The most recent comparable observations end in 2020 and rest on expert surveys. A financing constraint will also not necessarily appear as a high observed rate. It may appear as less debt offered, as projects that win support and do not reach financial close, or as the strike price a project needs in order to bid. A recurring harmonised survey of project financing terms would allow the question to be answered, by showing whether the convergence in sovereign and corporate borrowing has reached the projects that have to finance the transition.

Which financing functions still have to be performed

The historical evidence identifies several financing functions that the earlier arrangements combined. They supplied capital, they provided a route to cost recovery, and in each case they allowed new investment to draw on an established utility, customer base or contracted group of offtakers. Liberalisation separated functions that had previously been combined within those arrangements. The instruments that followed address them separately, through grants, contracts, loans, guarantees and other project-specific mechanisms.

Each acts on a different quantity. A grant reduces the amount of private capital that has to be raised. A contract for difference reduces uncertainty over future revenue. Public or concessional lending changes the price, the maturity or the availability of debt. A guarantee transfers specified risks, and can allow lenders to provide more debt or to provide it on better terms. These instruments act on different financing quantities and can therefore be used together. None by itself reproduces all of the functions documented in the earlier arrangements.

The evidence does not identify a single preferred instrument. It identifies the financing conditions on which different instruments act. A grant reduces the amount of capital a project must raise. Public lending changes the source, the price or the maturity of its debt. A long-term contract changes its exposure to future revenue

uncertainty. A guarantee transfers specified risks and may change the amount or the terms of the debt available. Comparing these instruments therefore requires attention to the financing condition each one changes as well as to the nominal amount of support.

Completing Europe's power transition therefore presents a different financing problem from building its first electricity systems. Member states enter the next investment cycle with different inherited fleets and, on the available evidence, different conditions when finance reaches the project, and those that entered the period after 2010 with smaller capital stocks have accumulated capital faster since. The higher general level of borrowing costs since 2018 and 2019 raises the burden on capital-intensive investment, while grants, contracts, public lending and guarantees act on different parts of that burden. Completing the transition therefore depends on the quantity of capital available, on the terms on which projects obtain it, and on the allocation of the risks that set those terms. Whether comparable projects can obtain comparable financing conditions across member states remains unresolved, because the project-level evidence is incomplete. A common market for electricity does not by itself create a common market for financing the assets that produce it.

Appendix

The body of the paper carries the evidence needed to follow the argument. This appendix carries the numbers needed to check it: every country, every variable, and the source and construction of each.

A. Generation and capital stock

Table A1	Generation systems, thirty countries
Table A2	The inherited capital stock, and where capital formation went

B. Prices and post-liberalisation support

Table B1	What consumers pay for the regulated network
Table B2	Post-liberalisation support and revenue-risk allocation

C. Liberalisation and historical financing

Table C1	Liberalisation and ownership over time
Table C2	How the first fleet was financed, and how its cost was recovered

D. Financing and technology assumptions

Table D1	The borrowing-cost ladder
Table D2	Technology assumptions behind the levelised cost figures

Each table begins on its own page. Tables that run to more than one page repeat their column headings and say so at the foot; where a table divides into panels, the panel is the unit to read.

Appendix A. Generation and capital stock

Table A1. Electricity generation by source, European countries, 2025

Every generation share quoted in Section 1 is drawn from here: all thirty countries and all six sources, rather than the five highest of each mapped in Figure 1.

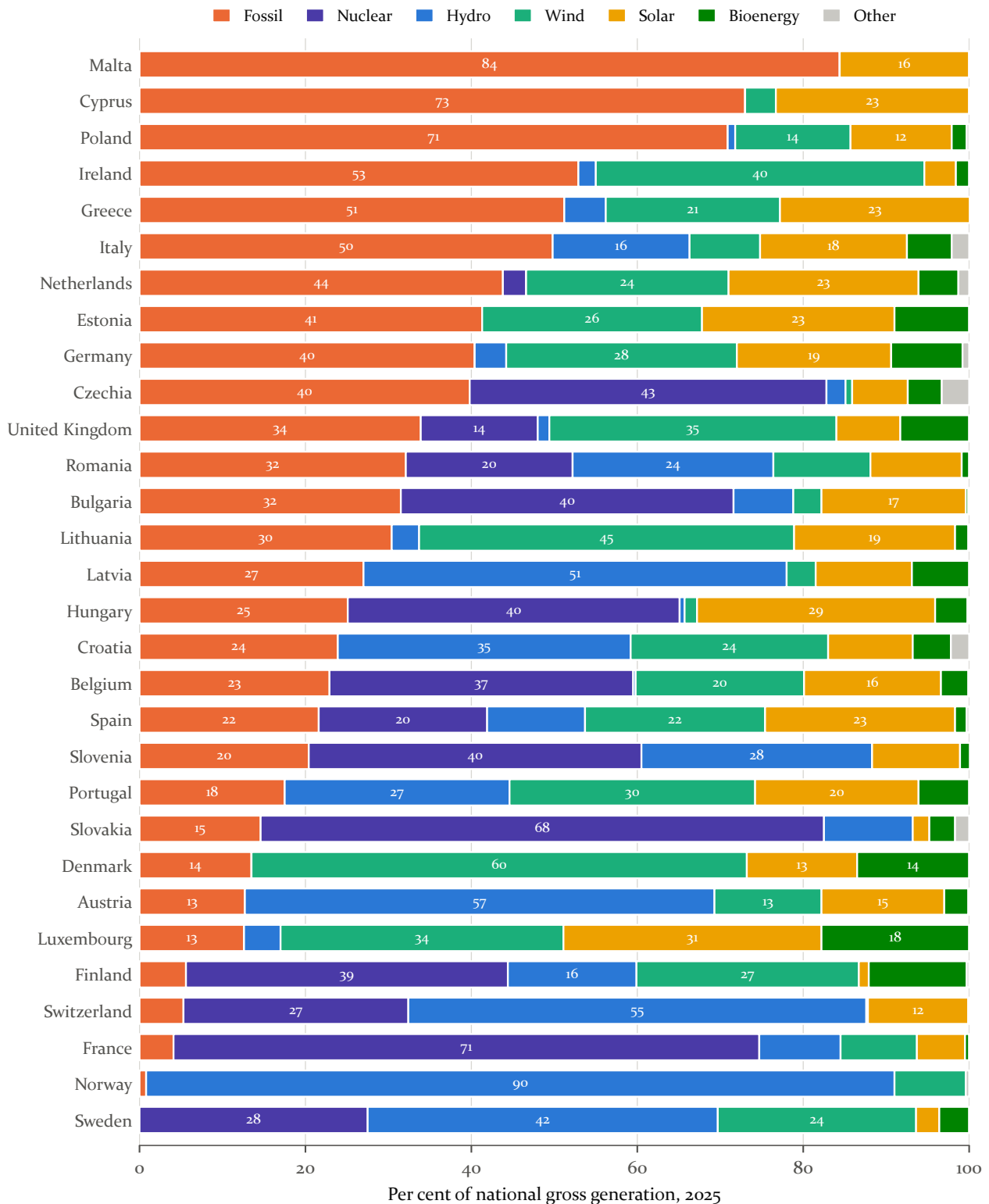


Figure 8. Thirty European power systems, each drawn as the shares of its own gross generation in 2025, sorted by fossil share. The bars compare composition and say nothing about size; total generation in terawatt hours is the first column of the table below, and it runs from 1.4 in Luxembourg to 526.6 in France. Shares are per cent of national gross generation. Values are printed inside a segment only where it reaches 12 per cent, and the full set is in the table below. Other collects the small residual left by Ember's other-renewables category and by rounding, at most 3.3 percentage points in Czechia. Sweden's fossil share is shown as zero because Ember reports no fossil row for it; the latest comparable Eurostat reading is 1.2 per cent.

Total generation runs from 1.4 terawatt hours in Luxembourg to 526.6 in France, and the twenty-seven member states generated 2,608 in 2025. The country-level shares behind the figure are in the replication files.

Source and construction

Ember, Monthly Electricity Data, downloaded 6 August 2026 to the monthly full release, licensed CC BY 4.0. Ember compiles the European country series from ENTSO-E and Eurostat and republishes it on one definition, which is the reason to use it rather than assembling the two by hand. Monthly values summed to the calendar year, and a country-year is dropped unless it carries twelve months. The extract covers 39 countries, 2010 to 2025.

Reliability, and what the cross-check against the official statistic shows

Eurostat energy balances, gross electricity production, was pulled for 2024 and compared country by country. Saved to the Eurostat cross-check.

Shares agree, levels do not. The fossil share differs by one percentage point or less for Germany, Italy, France, Czechia and Romania. The totals differ by up to 18 per cent, always with Eurostat higher: Portugal 18.5 per cent, Greece 15.0, Austria 13.6, Spain 13.5, France 9.5, Germany 9.2. Eurostat counts gross production including industrial autoproducers and small distributed plant, while the ENTSO-E series behind Ember covers metered units above a threshold. The Ember series is therefore suitable for the generation shares used here, but its terawatt hour levels should not be treated as interchangeable with Eurostat's.

Four countries where the fuel split diverges. Denmark 18.4 per cent fossil in Ember against 10.3 in Eurostat, because the two classify combined heat and power fuel differently. Spain 19.2 against 22.8, Belgium 26.4 against 23.0, Austria 10.0 against 13.3. Anything resting on these four should be checked in Eurostat first.

Absent is not zero. Ember omits a fuel row rather than writing a zero. Twenty-seven of thirty-eight European countries have at least one absent row, and most are genuine structural zeros, no nuclear in Austria, no coal in Norway. Two are not: Sweden has no coal or other-fossil row and shows zero fossil against Eurostat's 1.2 per cent, and Switzerland is missing coal, gas and bioenergy. Both understate fossil.

Which to cite. Eurostat for a paper table covering EU member states, since it is the official statistic and its 2024 data is final. Ember for 2025, which Eurostat has not yet published, and for Norway, Switzerland, the United Kingdom and the Western Balkans. Say which one each number came from.

Table A2. Gross fixed assets in the energy sector, 2023

Panel A gives the stock itself, Panel B the two periods compared, Panel C the series behind Figure 4. Notes and sources follow Panel A; the coverage and measurement cautions follow Panel C.

The table above measures a flow, one year of gross fixed capital formation, and it answers how fast a country is building now. It cannot answer what a country already owns. A fleet built in the 1970s and paid for decades ago barely appears in a 2023 flow, which is why France sits in the middle of the table above at €51.4 per megawatt hour and looks unremarkable beside Slovenia at €31.4. This table asks the other question, using the stock of fixed assets held by the same NACE sector D.

Gross rather than net is the right column for the question, because gross values the assets at what it would cost to replace them today and does not deduct for age. A country that built early is not penalised for having built early. Net fixed assets are shown beside it because the ratio of the two is a reading of how much of that replacement cost has already been written off, which is as close as the national accounts come to stating the age of a fleet.

France is the case that shows both what the measure adds and where it stops. In absolute terms France holds €304bn of sector D assets against Slovenia's €17.6bn, a factor of seventeen. Per megawatt hour produced the figures are €579 and €1,105. Part of that gap is real, because nuclear is expensive to build per kilowatt and cheap per megawatt hour delivered, running at a high capacity factor for most of the year, and because the same grid carries far more energy in France than in Slovenia. Part of it is measurement, for the reason set out in the last caution below. I therefore report the level and do not rank France against Slovenia on how much capital each holds.

Of France's sector D gross stock, 33 per cent is other buildings and structures and 61 per cent is machinery and equipment. In Germany the same split is 79 and 20, in Italy 84 and 15. A stock dominated by structures is a stock of networks, dams and transmission lines; a stock dominated by machinery is a stock of power stations. The split separates a country that has capitalised its grid from one that has capitalised its plant, and it does so more cleanly than any total, because it is a ratio within a country and the service-life problem below applies to both halves at once.

Panel A. The stock, 2023

Country	Gross fixed assets, €m	Net fixed assets, €m	Net as share of gross	Produced, TWh	Gross €/MWh produced	Gross stock ÷ 2023 investment, years
Luxembourg	10,123	6,141	0.61	2.5	4,049	50.3
Denmark	107,988	53,051	0.49	34.1	3,167	22.4
Lithuania	15,979	10,064	0.63	6.0	2,663	11.0
Slovakia	63,920	30,851	0.48	29.9	2,138	46.1
Italy	543,818	285,059	0.52	264.7	2,054	34.3
Estonia	11,531	8,313	0.72	5.7	2,023	15.0
Latvia	12,068	6,952	0.58	6.4	1,886	10.5
Portugal	79,859	33,927	0.42	49.0	1,630	39.2
Ireland	47,034	22,052	0.47	31.9	1,474	23.4
Netherlands	162,241	102,896	0.63	121.6	1,334	14.0
Hungary	46,872	24,879	0.53	35.5	1,320	26.1
Germany	673,710	346,590	0.51	522.9	1,288	19.6
Croatia	19,620	13,910	0.71	17.5	1,121	24.6
Slovenia	17,576	8,713	0.50	15.9	1,105	35.2

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Country	Gross fixed assets, €m	Net fixed assets, €m	Net as share of gross	Produced, TWh	Gross €/MWh produced	Gross stock ÷ 2023 investment, years
Czechia	83,632	34,773	0.42	77.0	1,086	21.9
Poland	180,139	51,773	0.29	167.4	1,076	23.7
Austria	79,905	43,186	0.54	75.0	1,065	16.1
Spain	295,558	189,610	0.64	285.9	1,034	26.8
Finland	75,117	44,956	0.60	81.5	922	19.7
Iceland	16,936	8,542	0.50	20.2	838	35.0
Belgium	69,795	40,388	0.58	83.7	834	18.1
Romania	45,998	44,185	0.96	58.0	793	17.7
Greece	36,652	17,076	0.47	49.9	734	14.3
Cyprus	3,294	2,355	0.72	5.3	622	10.5
France	303,992	173,013	0.57	524.9	579	11.3

Sorted by gross fixed assets per megawatt hour produced. Current replacement costs, million euro. The last column is gross stock divided by 2023 gross fixed capital formation, which is the average service life the country's own figures imply if its stock were in a steady state, and it is included as a test of whether two countries have measured their capital on the same basis.

Sources

Eurostat national accounts, stocks of fixed assets by industry, cross-classification of fixed assets by industry and by asset, NACE sector D, electricity, gas, steam and air conditioning supply, unit CRC_MEUR, current replacement costs in million euro. Asset codes NIIG total fixed assets gross and NIIN total fixed assets net. Electricity production is the same Eurostat energy balances series as the table above, so the two panels share a denominator.

Coverage and measurement

Gross and net stocks both come out of a perpetual inventory model whose service lives and retirement functions each country sets for itself, so a larger gross stock can be a longer assumed life rather than more plant. The last column is the test. Gross stock divided by one year of investment is the average service life a country's own figures imply if its stock were in a steady state, and it runs from 10.5 years in Cyprus, Latvia and France to 34.3 in Italy, 46.1 in Slovakia and 50.3 in Luxembourg. A growing stock biases the ratio down, and no growth path turns eleven years into thirty-four for two power systems built out of the same equipment. The level column therefore ranks countries only where the gap is large relative to that spread, which is why the France and Slovenia comparison above stops where it does. The growth rates below and in Figure 4 do not carry the problem, because a service-life assumption a country holds constant cancels in a within-country growth rate. The IMF takes the same view of national capital stock estimates and rebuilds its own on a single common depreciation rate for that reason.

Three narrower points. Six of the thirty-one countries in the panel above have no sector D stock at all, and the volume series is thinner again, for which see the series below. The numerator is the same broad sector as the panel above, holding gas networks and district heating alongside electricity against a denominator of electricity alone, which is most of why Denmark reaches €3,167. And a replacement cost is an accounting construct rather than a market value, so nothing here prices the risk on the assets, which is the subject of Table D1.

Panel B. The two periods, in numbers

Section 4 states the result and Figure 4 draws it; this is what sits behind both. The first table below is the summary the body used to carry. The growth rates are of the real stock in chain linked volumes, and the starting level is that stock divided by the electricity the sector produced in the starting year. The base is 2000 rather than 1995

because 1995 exists for seventeen countries and 2000 for all twenty-two, and the eight missing in 1995 are the central and eastern European systems the comparison is mostly about.

Real growth of the sector D capital stock	2000 to 2010	2010 to 2023
Correlation with the log of starting stock per megawatt hour	-0.07	-0.53
Slope, percentage points of annual growth per log point	-0.35	-1.64
Countries growing faster than 3 per cent a year	8 of 22	10 of 22
Countries growing slower than 1 per cent a year	7 of 22	4 of 22
Standard deviation of the log of stock per megawatt hour, at the end	0.49	0.45
Unweighted mean growth, per cent a year	2.69	2.39

Conditioning on a fixed base rather than the start of each period gives:

	2000 to 2010	2010 to 2023
Correlation with the log of the 2000 level	-0.07	-0.42
Slope on the 2000 level	-0.35	-1.19
SD of log stock per MWh at the end of the period	0.492	0.452
Highest country to lowest, at the end of the period	6.5	5.9

Conditioning on the 2000 level rather than the start of each period weakens the second-period relationship from -0.53 to -0.42 and does not remove it. Dispersion narrowed in both periods, from a standard deviation of 0.530 in 2000, so it is consistent with the claim and does not establish it; the conditional result is what separates the periods, because a level per megawatt hour can narrow through the denominator while investment goes wherever it likes.

The first question to put to a relationship of this shape is whether it is arithmetic. A small stock can post a high percentage growth rate from a small base, and a mature system grows slowly, so a negative slope could be mechanical. Four tests say it is not.

	2000 to 2010	2010 to 2023
Growth on the log of starting stock per MWh	-0.07	-0.53
Growth on the log of system size, terawatt hours	-0.44	-0.24
Growth on the log of the total stock	-0.48	-0.43
Growth on the log of the 2000 level per MWh	-0.07	-0.42

The first row is the test that matters most. A mechanical small-base effect, like regression to the mean and like measurement error in the starting level, operates in both windows by construction. It cannot appear in the second and not the first. The second row shows that the size effect exists and runs the wrong way for the objection, being stronger in the period where the stock relationship is absent. The fourth row is the standard correction for measurement error in the conditioning variable, using a level fixed ten years before the window opens, and it leaves the relationship at -0.42 on a slope of -1.19.

Entering the two together settles it. Regressing second-period growth on both the starting stock per megawatt hour and the log of system size gives the level -1.95, with a standard error of 0.55 and a t statistic of -3.56, and size -0.41, with a standard error of 0.18 and a t statistic of -2.24. Both are present and the intensity term is the larger of the two.

The second-period result is not carried by any one country. Removing each of the twenty-two in turn leaves the correlation between -0.42 and -0.64. Dropping the five systems producing under ten terawatt hours, where a

small denominator moves the level a long way, strengthens it to -0.74 on a slope of -2.20 , and the first period on the same seventeen countries reads $+0.11$.

Three cautions belong with the claim. Germany at $+0.46$ per cent a year in the second period is third from the bottom, which reads oddly beside the Energiewende until one remembers that household rooftop solar sits outside NACE sector D, so the series misses the channel that built a large part of the German fleet. Convergence in the physical stock is not convergence in what that stock cost to finance, and Table D1 shows the price of capital moving the other way over the same period. And a country can close the gap because the numerator rose or because the denominator fell, which is the case in Lithuania, where production fell from 11.4 terawatt hours in 2000 to 6.0 in 2023 after Ignalina closed.

Panel C. The series behind Figure 4

Real gross fixed assets in NACE sector D per megawatt hour produced, chain linked volumes in 2020 euro, and the annual real growth of the stock itself. These are the numbers the scatterplot in Section 4 is drawn from.

Country	€/MWh 2000	€/MWh 2010	€/MWh 2023	Growth 2000–10	Growth 2010–23
Austria	938	843	912	+0.43	+1.02
Belgium	509	455	653	+0.07	+1.86
Croatia	696	690	903	+2.74	+3.37
Cyprus	472	522	689	+5.74	+2.16
Estonia	234	366	1,575	+9.07	+5.09
Finland	609	538	761	+0.18	+2.80
France	263	303	515	+1.98	+3.51
Germany	739	724	930	+0.73	+0.46
Greece	418	465	651	+1.73	+1.52
Hungary	687	773	888	+1.80	+0.69
Ireland	548	980	1,150	+7.80	+2.14
Italy	1,098	1,484	1,785	+3.97	+0.40
Latvia	960	954	1,569	+4.76	+3.61
Lithuania	756	1,345	2,107	-1.10	+3.83
Luxembourg	2,531	1,092	3,046	+5.43	+3.41
Netherlands	923	748	1,135	+0.77	+3.41
Poland	571	648	908	+2.11	+3.10
Portugal	869	1,054	1,333	+4.13	+1.06
Romania	361	299	576	-0.19	+4.75
Slovakia	1,381	1,950	1,686	+2.36	-0.57
Slovenia	745	697	856	+1.22	+1.33
Spain	522	541	897	+3.36	+3.53

Twenty-two of the thirty-one countries in Table A2 have this series. Czechia, Denmark and Iceland report the stock at current replacement cost but not as a volume; Bulgaria's volume series stops in 2022; and Albania, North Macedonia, Norway, Serbia and Sweden report no sector D breakdown of fixed assets at all. Norway and Sweden are the costly absences, because an old and fully built hydro fleet is the case this measure would say most about.

Appendix B. Prices and post-liberalisation support

Table B1. What each country pays for its electricity network, 2024

No complete free source gives grid investment by country: Eurostat does not separate the network from generation inside the sector, the regulatory asset base annexes are unpublished for 2025, and the most recent transmission analysis is missing ten national operators. This table therefore measures the other end of the same pipe.

A complete free source is available for the other end of the same pipe. A regulated network recovers its capital and operating cost through a charge on the bill, so charge revenue measures what consumers pay for the grid in a year. It is a flow of payments over twelve months rather than a stock of assets or a year's capital spending. Over the life of the assets it is what investment costs, because a regulatory asset base is designed so that the charge repays the investment with an allowed return.

Germany's consumers paid €33.8bn for their network in 2024, France's €16.8bn, Spain's €10.4bn and Italy's €10.1bn. Measured against the size of the economy the order changes completely: Czechia at 1.02 per cent of GDP and Hungary at 1.00 sit at the top, Germany at 0.77 in the middle, and Ireland at 0.26 and Denmark at 0.35 at the bottom.

Country	Network revenue, €m	€ per MWh	Per cent of GDP	Household charge, €/kWh	Non-household charge, €/kWh
Czechia	3,271	62.8	1.02	0.0797	0.0556
Hungary	2,058	51.9	1.00	0.0578	0.0491
Finland	2,549	34.8	0.92	0.0620	0.0231
Latvia	347	56.0	0.85	0.1031	0.0376
Poland	7,238	53.5	0.85	0.0678	0.0490
Sweden	4,507	39.1	0.80	0.0622	0.0269
Lithuania	631	60.1	0.80	0.0756	0.0531
Germany	33,826	75.2	0.77	0.1217	0.0558
Estonia	297	45.2	0.74	0.0593	0.0374
Croatia	632	38.9	0.74	0.0474	0.0327
Bulgaria	765	25.1	0.73	0.0328	0.0194
Slovenia	444	37.0	0.66	0.0598	0.0245
Spain	10,427	47.5	0.65	0.0823	0.0294
Austria	3,095	53.4	0.63	0.0819	0.0362
Slovakia	797	38.9	0.61	0.0376	0.0394
France	16,803	44.6	0.57	0.0655	0.0306
Malta	123	44.0	0.53	0.0440	0.0440
Portugal	1,536	31.7	0.53	0.0458	0.0255
Netherlands	5,822	63.1	0.52	0.1254	0.0440
Greece	1,209	26.0	0.51	0.0400	0.0177
Norway	2,347	21.3	0.51	0.0393	0.0114
Romania	1,791	43.4	0.51	0.0654	0.0322
Italy	10,132	37.8	0.46	0.0623	0.0298
Belgium	2,848	39.5	0.46	0.0895	0.0249
Cyprus	143	28.3	0.41	0.0349	0.0244
Denmark	1,408	48.9	0.35	0.0799	0.0316
Ireland	1,482	46.4	0.26	0.0766	0.0357

Sources

Network charge. Eurostat household electricity price components and Eurostat non-household electricity price components, the network cost component of the electricity price, all-bands, euro per kWh, for household and non-household consumers. Extracted to the price components extract.

Consumption. Eurostat energy balances in GWh, siec E7000: final consumption by households, by industry and by commercial and public services. Services are charged at the non-household tariff.

GDP. Eurostat national accounts, gross domestic product, current prices, million euro.

What the measure is

A lower bound on network revenue, since transport consumption is excluded and any cost recovered through connection or capacity charges rather than the per-kilowatt-hour charge is outside the published price component. It is revenue and not investment: a country replacing a depreciated network shows a low charge while investing heavily, and the two diverge for years at a time. 2022 and 2023 are unusable here, because several member states cut network charges during the price crisis, so 2024 is the first clean year.

Table B2. Post-liberalisation support and revenue-risk allocation

This table reports three features of the post-liberalisation regime for each country: the reform year, direct government capital formation as a share of sector investment, and the remuneration instrument most often awarded in renewable auctions between 2011 and 2021. The last column translates that instrument into the revenue-risk allocation it implies. How the first fleet was financed is Section 2 and Table C2; the regulation index by country and year, and the state shareholdings, are Table C1. None of that is repeated here.

The reform year is dated from the data rather than from a policy narrative. It is the year of the largest single-year fall in the regulation index. Austria breaks first in 1988, then the United Kingdom and Norway in 1991, a cluster from 1998 to 2003 as the single market directives bite, and Slovenia last in 2007.

The revenue side is measured rather than inferred. Eight countries most often awarded a two-way contract for difference, which puts the state on both sides of the price, five most often awarded a sliding premium, which leaves the consumer paying one way only, and the rest sit on certificates, grants, another instrument or no auction in the window. That instrument does not describe every financing arrangement in a country, only the one most often awarded in auctions in that window, which is why the last column is stated as an implication and reads “not established here” where no auction was observed.

Country	Reform year	Government's own investment, % of sector	Instrument most awarded, 2011–21	Revenue risk implied by that instrument
Austria	1988	0.4	no auction in window	not established here
Norway	1991	0.2	no auction in window	not established here
United Kingdom	1991	n.a.	two-way contract for difference	state, both directions
Finland	1995	0.0	other	merchant or corporate offtaker
Sweden	1996	0.6	no auction in window	not established here
Spain	1998	0.7	investment grant	taxpayer at construction
Germany	1998	1.4	sliding premium	consumer, one direction
Estonia	1998	6.7	sliding premium	consumer, one direction
Switzerland	1998	n.a.	no auction in window	not established here
Denmark	1999	0.0	fixed premium	consumer and equity shared
Netherlands	1999	3.9	other	merchant or corporate offtaker
Portugal	2000	0.5	two-way contract for difference	state, both directions
Luxembourg	2000	5.6	sliding premium	consumer, one direction
Poland	2000	6.1	two-way contract for difference	state, both directions

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Country	Reform year	Government's own investment, % of sector	Instrument most awarded, 2011–21	Revenue risk implied by that instrument
Ireland	2000	7.6	two-way contract for difference	state, both directions
Italy	2000	1.6	two-way contract for difference	state, both directions
France	2002	7.9	two-way contract for difference	state, both directions
Slovakia	2002	9.4	sliding premium	consumer, one direction
Hungary	2002	0.7	two-way contract for difference	state, both directions
Greece	2002	1.1	two-way contract for difference	state, both directions
Czechia	2003	2.2	no auction in window	not established here
Belgium	2006	0.2	no auction in window	not established here
Slovenia	2007	10.5	sliding premium	consumer, one direction

Two columns are derived and the rule is stated rather than left implicit. Equity follows the state shareholding: fifty per cent or more reads as a state-owned balance sheet, above zero as mixed, zero as private. Revenue risk follows the instrument most often awarded in that country's auctions.

The network column

Across every country here the grid is financed the same way: the operator raises money from retained earnings and bonds, and a regulated charge recovers the cost over the life of the asset at a published allowed return. Table B1 gives what that charge collects, country by country.

State ownership over the period

State ownership of the largest generator rose between 2018 and 2023 in France, Poland, Ireland and Finland, fell only in Greece, and was unchanged in the other eighteen countries.

Sources

Regulation index and reform year. OECD Energy, Transport and Communications Regulation, electricity indicator, annual 1975 to 2023, in the ETCR extract.

State shareholding. OECD product market regulation sector database, question Q1a.1.6a_E1, government direct and indirect holding in the largest electricity generation firm on 1 January 2023. A shareholding recorded as “not applicable” means the government holds nothing and is read here as zero, not as missing.

Government's own investment. Eurostat government expenditure on fuel and energy capital formation against Eurostat NACE sector D gross fixed capital formation, mean of 2015 to 2024.

Instrument. AURES II auction database, Mendeley doi:10.17632/2h9jpy3wx.1, 820 rounds in twenty countries, 2011 to 2021, most frequently awarded remuneration scheme per country. “No auction in window” means that country held no auction inside the period the database covers, which is not the same as a merchant regime; Sweden and Norway ran certificate schemes instead and are genuinely market-exposed, while Austria, Czechia, Belgium and Switzerland are a coverage gap.

Appendix C. Liberalisation and historical financing

Table C1. Liberalisation and ownership over time, 1975 to 2023

Two different things are shown side by side because they are routinely confused. The index is liberalisation: entry barriers, vertical integration, market structure and public ownership combined into one score from zero to six, where six is a closed monopoly. The last two columns are ownership on its own. A country can move a long way on the first without moving at all on the second, and several did.

The shape

European electricity liberalisation was a fifteen-year event that finished around 2010. The mean index across these twenty-three countries sat at 5.80 in 1975 and had barely moved by 1990. It then fell 4.28 points, of which 93.5 per cent was complete by 2010 and 61.5 per cent happened in the single decade from 1995 to 2005. Between 2010 and 2023 it moved 0.28 points in thirteen years. Figure 3 plots the series. That matters for how the rest of the paper is built. The regime a country was left with in 2010 is the regime it still has, so it can be treated as a fixed characteristic of the country when explaining what was built afterwards, rather than as something moving alongside the investment it is meant to explain.

And privatisation did not follow

Ownership is measured only for 2018 and 2023, because the OECD publishes the components of the index at those two dates and the annual series is a composite only. The table below carries the 2023 level and the changes over that period are given here. Over that period the state's holding in the largest generator rose in France by 5.5 points, Poland 3.5, Ireland 1.0 and Finland 0.4, fell only in Greece by 16.9, and was unchanged in the other eighteen. The ownership data therefore show no general post-2010 shift towards privatisation.

The clearest cases of the two moving apart are Sweden and Norway, which fell 4.75 points on the index while holding their largest generator at 100 per cent, and Germany, which fell 4.15 points from a starting position of 4.50 rather than 6.00 and holds 10 per cent.

Country	1990	2000	2010	2023	Fall 1990–2023	State share 2023
Netherlands	6.00	2.33	0.76	0.00	6.00	0.0
Portugal	6.00	3.45	1.63	0.62	5.38	0.0
Italy	6.00	3.42	0.88	0.72	5.28	23.6
Denmark	6.00	3.40	2.03	0.75	5.25	50.1
Spain	5.27	0.50	0.12	0.12	5.15	0.0
United Kingdom	4.81	0.54	0.25	0.00	4.81	0.0
Sweden	6.00	3.25	2.25	1.25	4.75	100.0
Norway	6.00	2.25	2.25	1.25	4.75	100.0
Slovakia	6.00	6.00	1.35	1.35	4.65	n.a.
Finland	5.87	2.27	1.01	1.27	4.60	51.2
Greece	6.00	5.38	2.89	1.43	4.57	34.1
Poland	6.00	3.88	1.48	1.47	4.53	60.9
Germany	4.50	0.62	0.25	0.35	4.15	10.0
Czechia	6.00	4.84	1.89	1.90	4.10	69.8
Hungary	6.00	3.88	0.75	2.00	4.00	100.0
Slovenia	6.00	5.25	2.25	2.12	3.88	100.0

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Country	1990	2000	2010	2023	Fall 1990–2023	State share 2023
Switzerland	6.00	4.00	2.17	2.17	3.83	100.0
Luxembourg	6.00	3.51	1.72	2.32	3.68	40.4
Ireland	6.00	4.44	2.65	2.68	3.32	96.5
Belgium	3.84	3.08	1.25	0.75	3.09	0.0
Estonia	6.00	5.33	4.33	3.00	3.00	100.0
France	6.00	4.96	3.44	3.03	2.97	89.0
Austria	4.53	3.59	1.27	1.92	2.61	51.0

Sources

Index. OECD Energy, Transport and Communications Regulation database, electricity sector indicator, annual 1975 to 2023, scale 0 to 6.

Ownership. OECD product market regulation sector database, government direct and indirect holding in the largest electricity generation firm on 1 January, for 2018 and 2023 only. A holding recorded as not applicable is read as zero.

What is not available. No annual public ownership series for electricity is published free, since the OECD releases the sub-components of the index at five-yearly review dates. The trajectory column is therefore liberalisation and the ownership columns are two snapshots, and reading the index as a privatisation series would be wrong. The French 89.0 per cent for 2023 is the position on 1 January, before the buyout of minority shareholders completed that June took it to 100.

Table C2. How the first fleet was financed, and how its cost was recovered

Section 2 documents six countries in detail, because six is as far as the sources reach at that level of detail. This table covers all thirty at a coarser grain by putting one question to each of them. Through what mechanism did the incumbent system raise its investment capital, and what revenue recovered it? Every family below is an answer to that question and to nothing else.

A. Utility balance sheet. The utility raised the capital against itself, from retained cash or from borrowing in its own name, and a regulated tariff charged to a captive franchise recovered it.

B. State-backed capital. The state supplied, channelled or guaranteed the capital, and a price set to cover cost serviced it. A sovereign guarantee counts here, so the family does not claim that every unit of the money began in the treasury.

C. Consumer prepayment. Consumers supplied the capital directly, through cooperative and municipal ownership and through advance collection, and pricing was limited to necessary cost.

D. Offtaker consortium. The plant's own customers owned it, supplied its capital and were contractually obliged to take its output at cost.

E. Plan allocation. Investment was allocated administratively through the planned economy rather than raised against the expected revenue of the electricity undertaking, and electricity prices were administratively set. The family ends with the plan, and the route the World Bank prescribed for what followed is on record: its 1998 strategy for the region names low cost recovery among the six defining problems of the power industry and commits its lending to competition in generation and supply, market-based pricing and the sale of power assets to private strategic investors (World Bank 1998), which is the market rung of Section I reached without the intermediate phase France and Finland passed through.

Panel A. Countries in family A

Regime	Country	1990 index	Evidence
A. Utility balance sheet	Austria	4.53	partly documented, IBRD TO-115, Verbund combine programme 1956-58
	Belgium	3.84	partly documented, EIB 1978 and 1980, EBES and Intercom borrowing
	Cyprus	n.a.	documented, IBRD TO-562a, 1967-69 sources of funds
	France	6.00	documented, Section 2
	Germany	4.50	partly documented, Agora 2019 on tariffs; capital route inferred
	Greece	6.00	partly documented, PPC own bond issue from 1957
	Italy	6.00	partly documented, IBRD TO-162a, two utilities 1957-60
	Luxembourg	6.00	partly documented, IBRD TO-185-b, instruments but no split
	Malta	n.a.	partly documented, IBRD TO-320a, guarantee not established
	Netherlands	6.00	partly documented, EC 1999/796, offtake and cost regulation
	Portugal	6.00	documented, IBRD TO-507a, 1966-69 company plan

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Regime	Country	1990 index	Evidence
	Spain	5.27	documented, EC 1999/797 on the tariff; Rubio-Varas and De la Torre 2016 on the capital

Panel B. Countries in families B to E

Regime	Country	1990 index	Evidence
B. State-backed capital	Croatia	n.a.	documented, IBRD 1961, for 1961
	Ireland	6.00	partly documented, Central Fund advances; Act not read
	Norway	6.00	documented, Section 2
	Slovenia	6.00	documented, IBRD 1961, for 1961
	Sweden	6.00	documented, Prop. 1986/87:87, budget appropriations and state guarantee
	United Kingdom	4.81	documented, Section 2
C. Consumer prepayment	Denmark	6.00	documented, Section 2
D. Offtaker consortium	Finland	5.87	documented, Section 2, Mankala era; 1950s hydro differed
	Switzerland	6.00	documented, BFE study 2015 and KKG 2022, the Partnerwerk
E. Plan allocation	Bulgaria	n.a.	partly documented, IBRD 11250, regime described, no ledger
	Czechia	6.00	partly documented, IMF WP/96/129, plan decided power plants
	Estonia	6.00	regime assignment
	Hungary	6.00	partly documented, IBRD 5927-HU, partly self-financing by 1986
	Latvia	n.a.	regime assignment
	Lithuania	n.a.	partly documented, IBRD 11867, regime described, no ledger
	Poland	6.00	partly documented, IBRD 8224-POL on investment approval
	Romania	n.a.	documented, IBRD 391a-RO, 1973-79 sector programme
	Slovakia	6.00	partly documented, IMF WP/96/129, plan decided power plants

The classifying rule, and what it deliberately ignores

The families answer one question and separate three things that are usually collapsed into one word: who owned the utility, where its capital came from, and what revenue recovered the cost. Section 2 argues that the second and third decide the fleet and the first mostly does not, so the classification must not read the financing route off the ownership. France is in A with EDF wholly in state hands, because EDF raised the money against itself. A state equity contribution or a guaranteed loan running alongside the dominant route does not move a country between families, and France took both.

The column therefore names the route the evidence best supports as the dominant or characteristic one for large generation investment under the pre-liberalisation regime, and the evidence column says how firmly. For Croatia and Slovenia that distinction matters, since what is established is the mechanism at the date examined rather than its share of everything built before 1990. Several of these countries used more than one route, and the last column says where a second route is known to have been present. For Italy it is now measured, at 17.5 and 24.6 per cent of two utilities' four-year programmes, which leaves Italy in A. For the others it is not: Cyprus and Malta used externally guaranteed lending, Greece and Portugal took state capital alongside their utilities' own borrowing, and Spain ran private utilities beside a state-owned one. Until those proportions are measured none of them is a reason to move a country from A to B.

Evidence

Thirteen cells are documented, fifteen partly documented and two regime assignments. The distinction sits in the table's own column because it decides how much weight the table carries.

Documented means a source states the route. It covers five of the six countries of Section 2, plus Croatia, Slovenia, Portugal, Cyprus and Romania from the appraisals below, and three cases settled here.

Spain. The two sides come from two sources. Commission Decision 1999/797 records a pre-liberalisation tariff calculated plant by plant on standardised costs, which "guaranteed cost coverage for electricity generating plants built in Spain". Rubio-Varas and De la Torre (2016) record the capital: US Eximbank loans for 45 per cent of the American content with guarantees that brought private banks in for a further 30, the Spanish industrial banks as intermediaries, and the 60 to 70 per cent of each plant that was local cost raised in pesetas, largely by bonds sold to the savings banks under obligatory investment coefficients. The borrowing was in the companies' own names, with no state guarantee recorded except the 1967 French Treasury loan for Vandellós I, and the 1983 moratorium then paid the companies for the stranded plant out of the tariff. The coal fleet of the state-owned ENDESA, which drew on INI capital, is not covered, and no sector-wide proportion is given.

Sweden. Documented from the government's own bill on the state power board, Proposition 1986/87:87, and sitting in B rather than A. Vattenfall was a state enterprise whose investment came from a budget appropriation, 2,845 million kronor in 1985/86, and which paid the budget back calculated depreciation and interest on the state capital in use at a set return requirement, 12.625 per cent on an average 16,626 million in 1985. Ringhals was built that way. Forsmark belonged to a company three-quarters Vattenfall's, which borrowed on the open market under a state guarantee the Riksdag capped at 11,982 million kronor. Five of the seven reactors commissioned in the 1980s, Sweden's peak decade, took one of those two routes. Sydkraft and the municipal and industrial companies financed themselves as companies, so state ownership is not what places Sweden: Norway and France also owned their generators and sit elsewhere.

Switzerland. Moves from an assignment in A to documented in D. A substantial part of Swiss large hydro is held in Partnerwerke, plant companies owned by several utilities and the local public authorities, whose owners must take the output in proportion to their shareholding and pay the annual costs pro rata, those costs including depreciation, interest on the borrowed capital and a defined profit on the share capital (EVU Partners for the Federal Office of Energy, 2015). The form dates to Kraftwerke Oberhasli in 1925 and built Grande Dixence. Gösgen and Leibstadt, 2.3 of the 3.4 GW of nuclear, are Partnerwerke too, and the Gösgen accounts show the shareholders reimbursing 680 million francs of annual costs in 2022 against 350 million of share capital and 3.2 billion of long-term liabilities (Kernkraftwerk Gösgen-Däniken AG 2022). That is the Mankala company under another name. Beznau and Mühleberg were built inside integrated utilities and are A, but by capacity D is the route. As with Finland the documents are present-day accounts of arrangements dating from each company's founding, which is the caveat on both.

Croatia and Slovenia. Both rest on the IBRD's 1961 appraisal of Yugoslav power projects, which records each generating plant as "normally constructed and operated as an individual enterprise" under an elected workers' council, the Investment Bank making "loans covering the entire estimated cost of the project" to those enterprises, and federally set rates "generally sufficient to cover all operating expenses". That is credit channelled through a state institution and repaid from a cost-covering price, which is family B rather than allocation under a command plan. The evidence is for 1961; later Yugoslav arrangements have not been checked.

Partly documented means the structure is sourced and the capital route is not, or the evidence is for one company or one date rather than the country and the period. It covers fifteen countries drawn from six kinds of source: the Bank archive for Italy, Luxembourg, Malta, Austria, Hungary, Bulgaria, Lithuania and Poland; the Commission's stranded-cost decisions for the Netherlands; an IMF working paper for Czechia and Slovakia, where Bulíř records that "construction and financing of major fixed capital ventures, such as CMEA investment projects, nuclear power plants, and the like, were still decided by the Planning Authority"; a company history for Greece, whose first bond loan of 90 million drachmae at 8 per cent was offered for subscription on 1 July 1957; the European Investment Bank's annual reports for Belgium, which lend Bfrs 2,500 million to EBES in 1978 and Bfrs 5,000 million to Intercom and Ebes in 1980, privately owned utilities borrowing long term in their own names; and the Oireachtas for Ireland. Germany belongs here too, since Agora Energiewende records the area monopolies and the approved tariffs but not how the utilities funded their investment, and no source read here supplies it.

Regime assignment is now only Estonia and Latvia, in family E. They were Soviet republics until 1991 and their plants were state property allocated through the planning system, but no source read here states that for either by name, neither has a sector document of its own, and the Lithuanian review that describes the Baltic system names only Lithuania. They are a classification rather than a finding, and either may move when a document is found. That is why this table is not drawn as a map: a categorical choropleth would give a documented cell and an assigned cell the same colour.

The 1990 column is context and not evidence for the classification. It does not separate A from B or either from the rest. Where the OECD series exists it shows these different routes running inside markets still largely closed to entry, since every country with a series stood at or near the top of the index in that year, the value for a closed monopoly with no third-party entry. Seven countries have no series at all: Cyprus, Malta, Croatia, Bulgaria, Latvia, Lithuania and Romania.

Panel C. The IBRD financing plans

IBRD project appraisals give comparable financing plans for thirteen historical cases across ten countries. The Bank lent to power projects across Europe through the 1950s and 1960s, the appraisals are public and in English, and each carries a sources of finance section, so one series read the same way for every country is better evidence than thirty national histories written to thirty different standards.

Borrower and scope	Report	Financing plan as the document gives it	What it settles
Italy, group programme 1957-60	IBRD TO-162a, 1958	115.23bn lire: 33.12 revenues, 22.28 share capital, 39.63 own borrowing, 20.21 Bank via the Cassa	82.5 per cent raised against the company. Italy to A, partly documented
Italy, company programme 1957-60	IBRD TO-162a, 1958	44.7bn lire: 14.2 revenues, 10.0 stock, 9.5 local borrowing, 11.0 Bank via the Cassa	75.4 per cent raised against the company, same direction
Italy, one nuclear plant 1959	IBRD T.O. 200a, 1959	\$66.40m: \$40m Bank via the Cassa, \$26.40m from fourteen shareholders, half stock half advances	Family D structure on state-channelled credit. One plant, so it moves nothing
Yugoslavia, two projects 1961	IBRD TO-267a, 1961	The Investment Bank relends its own and the Bank's resources covering the entire project cost to each enterprise	Croatia and Slovenia out of E and into B
Austria, one hydro project 1955	IBRD T.O. 75-c, 1955	1,003m schilling: 323 own resources, 197.2 an Austrian bank, 222.8 lent by its own consumers, 260 the Bank	Consumers lend and take output at cost to 2010, so family D. One company, so Austria stays A

continued on next page

Borrower and scope	Report	Financing plan as the document gives it	What it settles
Finland, five projects 1956	IBRD T.O. 105-a, 1956	No cost split. The Bank lends to the Mortgage Bank under a state guarantee, which relends to the companies	Family B for the 1950s hydro, not the Mankala era Section 2 documents
Romania, sector programme 1973-79	IBRD 391a-RO, 1974	\$4.4bn: 46 per cent internal cash, 52 state budget, 1 each the Investment Bank and the IBRD	Budget allocations adjust to whatever the plan needs. Romania to E, documented
Hungary, sector finances 1986	IBRD 5927-HU, 1986	40 per cent of depreciation to the state budget, 60 retained, plus a Development Fund from profits	Substantially enterprise-financed by 1986, which complicates E rather than confirming it
Portugal, company plan 1966-69	IBRD TO-507a, 1966	2,180.4m escudos: 895.4 own resources including 620.0 new share capital, 1,285.0 borrowings of which 575.0 the Bank	Raises against itself in the market. Portugal to A, documented
Cyprus, sources of funds 1967-69	IBRD TO-562a, 1967	£6,173k: 2,531 net internal cash, 1,455 drawdown of own balances, 320 customer contributions, 1,867 Bank	Finances the major part from its own cash. Cyprus to A, documented
Malta, capital programme to 1969/70	IBRD TO-320a, 1963	34 per cent internal cash, 6 government advances as subordinated debentures, 58 foreign borrowing, 2 local	One external loan dominates. A retained, marked partly documented
Luxembourg, one project 1958	IBRD TO-185-b, 1958	\$40m from a share capital increase, bonds in New York, Luxembourg and Germany, the EIB and the Bank at 22 per cent	Company equity and market bonds. A retained, partly documented
Bulgaria, regime description 1993	IBRD 11250, 1993	No historical ledger. Describes an economy organised for central planning within the CMEA bloc	Places the regime only. Bulgaria to E, partly documented

Two points belong with the table. Nothing in it converts a project financing plan into a national share, which is why Austria stays in A on the strength of one company and why the Italian cells say partly documented. And the method fails on Greece: the Bank's lending there runs to irrigation, groundwater, forestry, education and sewerage and returns no power project at all, which is why the Greek cell rests on the utility's own history instead.

The offtaker consortium of family D is not a Finnish peculiarity. Switzerland ran it from 1925 as the Partnerwerk, and it appears in Austria in 1955 and in Italy in 1959, before the Finnish courts settled the Mankala question in the 1960s. Europe reached for the same instrument wherever a plant was too large for one balance sheet and its output could be committed in advance.

The full extraction, with every component as the document states it, is in the replication files.

Appendix D. Financing and technology assumptions

Table D1. The borrowing-cost ladder, 2018 onward

Four measures of the price of money in the same country, ordered from the state outward to the power project. The first two rungs are averages of 2018 and 2019; the project rungs take every reported estimate with a financing year of 2018 or later, and the mismatch between the two windows is quantified below. The first two are prices at which transactions took place, reported to the ECB by banks. The last four are what practitioners told researchers they were seeing when surveyed. The two kinds of evidence are kept apart because the paper's argument depends on comparing them.

All figures are per cent except the debt share. The project columns pool every solar and onshore wind estimate for the country, which is what Figure 2 draws and what the paper quotes; the last two columns give the same quantity for each technology on its own, so the pooling can be checked. Pooling is used because the question is what a country's financing environment does to a project rather than what a technology costs, and because it leaves no country resting on a single estimate. The cost is that plant risk differs between the two.

Panel A. The ladder

	Sovereign	Corporate bank	Project debt	Equity return	Debt share	Pooled	Solar	Wind
Germany	0.07	1.59	2.01	4.82	81.9	1.93	1.83	2.04
Belgium	0.49	1.57	2.33	8.83	78.3	3.72		3.72
France	0.46	1.46	2.13	6.32	77.7	3.92	4.44	3.04
Finland	0.36	1.51	1.83	6.17	63.8	3.98		3.98
Netherlands	0.25	1.25	1.67	6.73	62.4	4.02	4.40	3.73
Slovenia	0.60	2.02	3.50	9.50	75.0	4.50		4.50
Italy	2.28	1.77	3.37	7.03	67.6	4.61	4.33	4.71
Spain	1.04	1.72	3.28	8.29	59.2	4.95	4.98	4.90
Slovakia	0.57	2.17	3.70	10.00	82.0	5.50	5.50	
Greece	3.39	4.03	3.69	10.00	75.0	5.50	5.60	5.41
Austria	0.37	1.48	3.20	7.79	74.6	5.59	6.11	5.33
Portugal	1.30	2.43	3.42	9.17	53.1	6.18	6.82	5.22
Croatia	1.73		4.59	10.12	71.7	6.52	5.77	6.90
Bulgaria	0.66		4.22	12.83	70.0	6.70	7.30	5.50
Ireland	0.64	2.94	4.17	10.17	73.8	6.98	9.93	5.50
Estonia		2.91	3.68	12.76	71.0	7.63	7.21	8.06
Lithuania	0.31	2.70	2.45	8.00	72.0	8.75		8.75
Latvia	0.62	3.15	4.86	19.53	55.3	10.59	12.10	9.08
Cyprus	1.62	3.40						
Luxembourg	0.22	1.33						
Malta	1.03	3.80						

What a country burns and what its capital costs are close to unrelated. Across the eighteen countries with both, the fossil share of 2025 generation and the project cost of capital correlate 0.12. Latvia at 27 per cent fossil pays 10.59 per cent for project capital while Germany at 40 per cent pays 1.93, and Ireland at 53 per cent fossil pays 6.98 while Italy at 50 per cent pays 4.61.

Panel B. How much evidence stands behind each project figure

The first two columns of the table above are monthly ECB series, reported in every month of both years for every country shown. The project columns are not comparable to them in weight. Each is the mean of every nominal after-tax observation in Steffen et al. (2025) with a financing year of 2018 or later, and for several countries that mean rests on one observation from one study. The first count below is for solar and the second for onshore wind, the two technologies pooled in the project columns of the table above and in Figure 2.

Solar

	Obs	Financing years	Studies
Germany	4	2018, 2019, 2020	Egli et al. 2023; Schmidt et al. 2019; IRENA 2023
France	5	2018, 2019, 2020	Egli et al. 2023; Đukan and Kitzing 2023; IRENA 2023
Spain	4	2018, 2019, 2020	Đukan and Kitzing 2023; Vázquez-Vázquez et al. 2021; IRENA 2023
Greece	3	2018, 2019, 2020	Đukan and Kitzing 2023; IRENA 2023
Netherlands	3	2018, 2019, 2020	Đukan and Kitzing 2023; IRENA 2023
Portugal	3	2018, 2019, 2020	Đukan and Kitzing 2023; IRENA 2023
Bulgaria	2	2019, 2020	Đukan and Kitzing 2023; IRENA 2023
Latvia	2	2018, 2019	Đukan and Kitzing 2023, one estimate entered twice
Slovakia	2	2018, 2019	Đukan and Kitzing 2023
Austria	1	2020	IRENA 2023
Croatia	1	2020	IRENA 2023
Estonia	1	2019	Đukan and Kitzing 2023
Ireland	1	2020	IRENA 2023
Italy	1	2020	IRENA 2023

Onshore wind

	Obs	Financing years	Studies
Germany	4	2018, 2019, 2020	Schmidt et al. 2019; Đukan and Kitzing 2023; IRENA 2023
Netherlands	4	2018, 2019, 2020	Đukan and Kitzing 2023; IRENA 2023
Finland	3	2018, 2019, 2020	Đukan and Kitzing 2023; IRENA 2023

continued on next page

	Obs	Financing years	Studies
France	3	2018, 2019, 2020	Đukan and Kitzing 2023; IRENA 2023
Greece	3	2018, 2019, 2020	Đukan and Kitzing 2023; IRENA 2023
Italy	3	2018, 2019, 2020	Đukan and Kitzing 2023; IRENA 2023
Spain	3	2018, 2019, 2020	Đukan and Kitzing 2023; IRENA 2023
Austria	2	2018, 2019	Đukan and Kitzing 2023
Ireland	2	2019, 2020	Đukan and Kitzing 2023; IRENA 2023
Latvia	2	2018, 2019	Đukan and Kitzing 2023, one estimate entered twice
Lithuania	2	2018, 2019	Đukan and Kitzing 2023
Portugal	2	2018, 2019	Đukan and Kitzing 2023
Belgium	1	2019	Đukan and Kitzing 2023
Slovenia	1	2018	Đukan and Kitzing 2023

The wind ladder is the better supported of the two at the top and the thinner at the bottom. Germany again carries four observations from three methods, and seven countries carry three or four, against four countries in the solar count. At the other end Belgium and Slovenia rest on one survey estimate each, and Latvia repeats its 9.08 in both years as it does for solar. Austria, Lithuania and Portugal have two genuinely different years, which the solar count does not have for any country outside the IRENA studies.

Latvia remains the thinnest cell the paper leans on, and pooling is what keeps it usable. Its solar figure of 12.10 appears as two rows, 2018 and 2019, carrying identical values on all four variables, and its wind figure of 9.08 does the same, so each technology on its own averages one elicitation with itself. Pooled they are two elicitations rather than one, giving 10.59. Germany is the opposite case, with eight observations from four methods. The Germany against Latvia contrast is therefore a four-source blend set against two survey estimates, and is used as an illustration of magnitude rather than as a measured national difference. The fifteen-country ladder is the evidence for the widening itself.

The windows also differ. The ECB rungs average 2018 and 2019 while the project columns take everything from 2018 onward, which in practice runs to 2020. Recomputing the ECB rungs over 2018 to 2020 moves the Germany against Latvia sovereign gap from 0.55 to 0.52 points and the corporate gap from 1.56 to 1.64, and the fifteen-country sovereign range from 3.31 to 2.80. A matched window is not available, because Germany reports no project cost of debt in 2018 or 2019 and the benchmark would disappear. One data note: all 329 rows attributed to Đukan and Kitzing (2023) carry consecutive DOIs, which is a fill-down artefact in the spreadsheet; the first is the real one and no value is affected.

Panel C. Supplementary statistics

Three results the body states and this table documents. All are computed on the fifteen countries observed at all four rungs, being Austria, Belgium, Germany, Spain, Finland, France, Greece, Ireland, Italy, Latvia, Lithuania, the Netherlands, Portugal, Slovenia and Slovakia, except the decomposition, which runs against Germany for the seventeen countries with a project cost of capital.

Per cent, except the range	Sovereign	Corporate bank	Project debt	Project WACC
Mean	0.85	2.12	3.04	5.38
Standard deviation	0.88	0.79	0.93	2.14
Range, percentage points	3.31	2.79	3.18	8.66

Countries lie 2.7 times further apart at the project cost of capital than at the corporate bank stage and 2.4 times further apart than at the sovereign stage; by range, 3.1 and 2.6 times. The corporate stage is the least dispersed of the four, below even the sovereign. Dispersion is in percentage points rather than as a coefficient of variation, because Germany's sovereign rate of 0.07 puts a number close to zero in the denominator.

The gap against Germany, split by Shapley decomposition over the three ways a blend can move. Writing the vanilla blend as $L \cdot k_d + (1-L) \cdot k_e$, each factor takes its average marginal contribution across all six orders in which the three can be switched from the German value, so the order of switching does not decide the answer. Percentage points.

	Gap	Gearing	Equity return	Loan rate	Of which sovereign
Latvia	8.89	2.32	4.62	1.95	0.45
Bulgaria	4.29	0.68	1.93	1.68	0.48
Estonia	3.80	0.65	1.87	1.28	
Croatia	3.64	0.43	1.23	1.98	1.36
Portugal	3.60	1.23	1.41	0.95	1.00
Ireland	3.23	0.36	1.19	1.69	0.47
Spain	2.80	0.89	1.02	0.90	0.79
Greece	2.75	0.32	1.12	1.32	2.71
Slovenia	2.48	0.30	1.01	1.17	0.43
Slovakia	2.32	-0.00	0.93	1.39	0.40
Italy	2.04	0.46	0.56	1.02	1.81
Austria	1.85	0.27	0.65	0.93	0.25
Lithuania	1.49	0.41	0.73	0.34	0.19
Belgium	1.22	0.17	0.80	0.26	0.35
Netherlands	1.06	0.77	0.53	-0.24	0.15
Finland	0.88	0.65	0.37	-0.13	0.24
France	0.54	0.15	0.30	0.10	0.32

Pooled across the thirteen, the gap is 39 per cent required equity return, 38 per cent loan rate and 23 per cent gearing, with the sovereign carrying 23 per cent of the whole. The last column is what the sovereign spread over Germany would contribute at German gearing, and Greece is the case that settles the sovereign question: 2.84 points predicted against a measured total gap of 2.33, so Greek solar projects were financed on better terms than the Greek state was. Across the twenty countries with both series the sovereign rate and the project cost of capital correlate 0.19 on a slope of 0.33, which is weak and is a cross-sectional association rather than a pass-through coefficient.

What the terms do track is how much a country has already built. Against the log of cumulative wind and solar capacity in 2018, for the fourteen countries with both, the required equity return correlates -0.86, the project cost of capital -0.75, the project loan rate -0.31 and the debt share +0.26. So the two terms that drive the gap have different origins and only one of them is explained here. Causation runs both ways, since cheap capital also produces deployment.

Panel D. The German benchmark

Germany is the comparator in every column, so its 1.93 per cent carries more weight than any other figure. It is the mean of eight nominal after-tax observations from 2018 onward, three for solar and five for onshore wind, and Germany is the only country here whose observations come from studies other than the expert surveys that supply the component columns. Egli et al. (2023) re-engineer winning auction bids and report 2.10 and 1.50 for solar in 2019 and 2020; Schmidt et al. (2019) combine market rates with estimated mark-ups and report 1.55 for solar and 1.89 for wind in 2018; Đukan and Kitzing (2023) report 1.90 and 1.92 for wind in 2018 and 2019; IRENA (2023) surveyed experts and reports 2.16 for solar and 2.44 for wind in 2020, and the component columns above are that survey and Đukan and Kitzing together.

A reader will notice that 1.93 sits below both the German cost of debt of 2.01 and the required equity return of 4.82, which no blend of the two can produce. The two are measured on different bases: at the survey's own German values the pre-tax blend exceeds the after-tax figure the same survey reports, so the interest deduction accounts for about half a point, and 1.93 is lower again because it averages the surveys with auction and market estimates. The pooled figure is used throughout because the same rule is applied to every country.

Why the level of interest rates cannot identify the mechanism

Section 8 compares what France and the United Kingdom built, and notes that the level of international interest rates cannot account for the contrast. This is the estimate behind that statement, kept here because it establishes a limit rather than a result.

A large power station appears in the capacity statistics several years after its financing was arranged, so the rate prevailing at commissioning measures the wrong year. Regressing the capital-intensive share of capacity commissioned in a country and year, from the Joint Research Centre vintages, on the real long-term rate averaged over the window five to ten years earlier gives a coefficient of -0.029 with a t statistic of -3.0 , over fourteen countries and 305 country-years from 1965 to 2015, with country fixed effects and standard errors clustered by country.

Adding year fixed effects removes it, and that is the informative part. Interest rate variation over this period is largely common across Europe, so once the common movement is absorbed there is little independent variation left in the regressor. An equation that could identify the effect would have to interact the rate with the extent to which a country's financing arrangement absorbed the revenue risk, which requires a measure of that arrangement the paper does not have. The cross-country comparisons in Section 8 illustrate the mechanism and do not estimate it.

Sources and cautions

Government borrowing: ECB, long-term interest rate for convergence purposes, ten-year maturity. Corporate bank borrowing: ECB, Monetary Financial Institutions Interest Rate Statistics, composite cost of borrowing for non-financial corporations. Project columns from Steffen et al. (2025), solar PV, financing years 2018 onward, nominal after-tax only, with the underlying Latvian, Greek and most other 2018 to 2019 observations attributed within that database to Đukan and Kitzing (2023), whose European observations come from the survey reported in AURES II Deliverable 5.2.

The ECB series cover euro area members only, so Poland, Hungary, Czechia and Romania appear in Figure 5 with a project cost of capital and have no sovereign or corporate entry here.

The ECB series for large loans with the interest rate fixed beyond ten years, which is closer in tenor to a power project than the composite is, is too thin in small countries to use. Latvia reports it in three months of 2018, five of 2019 and four of 2025, three of the last at exactly 18.00 per cent. The composite is reported in all twelve months of every year and is the series used above. The replication files record the number of reporting months behind every cell.

The project columns are survey estimates and several rest on one observation. Ireland, Austria, Croatia and Italy have a single observation each. Latvia has one expert survey entered twice, for 2018 and 2019, with identical values on all four variables. AURES II says of Latvia, Slovakia and Cyprus that they “present very little development of wind onshore capacity in the period 2017 to 2019, hence, the values may not fully depict the real spread between minimum and maximum WACC values”.

The components are separately averaged in the source, so multiplying them does not reproduce the reported blend. For Latvia the components give 11.41 against the 10.59 reported. The decomposition above therefore reconstructs the mechanism rather than the study's own arithmetic.

Table D2. The technology assumptions behind Figures 6 and 7

Figures 6 and 7 are sensitivity calculations. They hold the plant fixed and vary only the cost of capital, so everything that moves in them is financing. This table is the plant.

The technology cases are a constructed benchmark and not a single published dataset. They combine 2024 IRENA observations for solar and onshore wind, deflated to 2018 dollars, with IEA/NEA benchmark assumptions for nuclear, coal and gas, expressed in common 2018 USD. Nothing here is a result reported by either organisation. The cost of capital at which nuclear passes gas in Figure 1b is an output of the model below applied to those inputs, not a finding of the IEA/NEA study.

Both sources publish in dollars, so no exchange rate enters and the figures are in 2018 USD throughout.

Panel A. Inputs

	Overnight capital cost (2018 USD per kW, net capacity, before grid connection)	Capacity factor (per cent of rated output over a year)	Operating life (years)	Construction period (years)	Operations and maintenance (2018 USD per MWh generated)	Fuel (2018 USD per MWh generated)	Carbon (2018 USD per MWh generated)
Solar PV, utility scale	553	17.4	25	1	6.97	0	0
Onshore wind	833	34.0	25	1	17.73	0	0
Nuclear	3 370	85.0	60	7	14.26	9.33	0
Coal	1 785	85.0	40	4	17.01	25.41	24.16
CCGT gas	955	85.0	30	3	6.99	45.50	10.10

Operations and maintenance is stated per MWh generated and includes both the fixed and the variable component, because that is how the source levelised cost tables report it. The three columns to its right are also per MWh generated rather than per MWh of fuel input, so they already carry the plant's conversion efficiency. Overnight capital cost excludes interest during construction, which the model adds, and excludes grid connection, which the IEA/NEA study places outside its boundary.

Where each input comes from

Solar and onshore wind, capital cost and capacity factor: IRENA, *Renewable Power Generation Costs in 2024*, Table SI, global weighted averages of USD 691 and 1,041 per kW at 17.4 and 34 per cent, in 2024 dollars.

Nuclear, coal and gas, capital cost: IEA/NEA, *Projected Costs of Generating Electricity*, 2020, Table 3.1, median overnight costs of USD 3,370, 1,785 and 955 per kW across each country's median case, in 2018 dollars.

Deflation. The two renewable rows are multiplied by 0.8005, the ratio of the US consumer price index in 2018 to 2024, giving 553 and 833. The step is load bearing rather than housekeeping, since undeflated the renewable rows would look a quarter cheaper than the thermal ones for a reason unconnected to technology, which alone moves solar past nuclear in the capital column.

Lives, construction periods and the thermal capacity factor are the study's harmonised assumptions (pp. 37 to 39); its 85 per cent is a convention for comparing plant that could run in baseload rather than a forecast.

Fuel and carbon are the study's European prices (p. 41): gas USD 8.00 per million Btu, hard coal USD 75.00 per tonne, carbon USD 30.00 per tonne of CO₂, nuclear fuel USD 9.33 per MWh. Gas and coal are converted

to a charge per MWh at the study's median European efficiencies of 60 and 42 per cent, using energy content and emission factors recovered from its own columns; all recover to standard values, which is the check that the columns were read the right way round.

Operations and maintenance comes from the levelised cost tables at a 7 per cent discount rate, each figure one named plant, because that spread carries no country label of its own.

Two limitations

The EGC contains no European coal plant. Its coal submissions are from Australia, Japan, Korea, the United States, Brazil, China and India, so the coal row is an OECD median priced at European fuel and carbon, and not a European plant.

The nuclear median is held down by Korea, China and India. The study's two European submissions are France at USD 4 013 per kW and the Slovak Republic at USD 6 920. Substituting them raises the capital column from 3.53 times gas to 4.20 and 7.25 respectively, so the nuclear result in Figure 6 is heavily dependent on the construction cost assumed and should be read as such.

Panel B. What the inputs imply

	Capital committed before operation (2018 USD per MWh of annual output)	Relative to CCGT gas	Capital share of levelised cost at 6 per cent (per cent)	Levelised cost at 3 per cent (2018 USD per MWh)	at 7 per cent	at 10 per cent
Solar PV, utility scale	363	2.83	81	28.9	39.5	48.4
Onshore wind	280	2.18	55	33.8	41.7	48.6
Nuclear	453	3.53	59	41.5	63.4	85.1
Coal	240	1.87	21	77.4	86.5	95.0
CCGT gas	128	1.00	14	69.3	73.7	77.6

The first two columns are the overnight capital cost divided by one year's output. They measure how much upfront investment has to be financed for a given amount of annual output, and they are not levelised costs of electricity. The paper quotes the second column rather than the first for that reason. The two answer different questions. Capital committed before operation asks how much money must be raised in advance. Levelised cost asks what construction, financing, operation, fuel and carbon together cost per MWh over the plant's life, and it depends on the operating life while the capital ratio does not.

The pattern the two columns together produce is the substitution the paper is about. Gas requires the least upfront capital and coal somewhat more, wind and solar substantially more and nuclear the most among these cases, while gas and coal go on paying substantial fuel and carbon charges for as long as they run.

Operating life is worth most when capital is cheap, which is the same mechanism seen from another side. Shortening the nuclear life from 60 years to 40 raises its levelised cost by 9.4 per cent at a 3 per cent cost of capital and by 1.5 per cent at 10 per cent, because output forty to sixty years out discounts to almost nothing. A high cost of capital does not only raise the cost of a long-lived asset, it removes the value of the long life.

Panel C. The model, and how closely it follows the source method

Levelised cost is the overnight capital cost carried forward to the commissioning date, divided by discounted lifetime output, plus operations, fuel and carbon per MWh. Two conventions are needed that the EGC does not state and that were recovered from it. Construction spending is linear over the construction period and treated as paid mid-year, which reproduces the study's own investment cost columns exactly. Output is discounted at mid-year, without which the rebuilt cost runs 0.6 per cent high at a 3 per cent discount rate and 3.9 per cent high at 10. Solar output falls 0.5 per cent a year, which the study applies and which this model applies too.

The model rebuilds the published levelised cost of seven named plants from the extracted inputs at 3, 7 and 10 per cent. The purpose is to establish that the model follows the source methodology closely rather than to claim precision, and the maximum error across all twenty-one comparisons is 0.8 per cent of the published figure.

Plant	Published at 7 per cent	Rebuilt	Difference
Denmark solar PV 8 MW	41.11	41.37	+0.26
Denmark onshore wind 4.48 MW	29.18	28.97	-0.21
France onshore wind 50 MW	56.08	56.07	-0.01
Austria onshore wind 3 MW	76.35	76.95	+0.60
Italy CCGT 790 MW	69.47	69.47	0.00
Korea coal 954 MW	75.59	75.59	0.00
France EPR 1 650 MW	71.10	71.10	0.00

Nominal and real

The EGC discount rates are real and the model above is in constant 2018 dollars. The project cost of capital in Steffen et al. (2025) is nominal after tax. Figure 5 uses the German and Latvian project estimates, so those two rates are converted before they are used.

The deflator is the euro area long-term inflation expectation that stood while those financing decisions were being made, rather than a policy target or a realised price index. It is the ECB Survey of Professional Forecasters, average of point forecasts for the longer term, four to five calendar years ahead, averaged across the eight quarterly rounds of 2018 and 2019, which gives 1.81 per cent. Germany's 1.93 per cent nominal becomes 0.12 per cent real and Latvia's 10.59 becomes 8.62.

One euro area expectation is applied to both countries rather than each country's own realised inflation. Both were euro area members over this window, so the euro area expectation is the one embedded in a financing decision in either, and using two different national price indices would mix a difference in financing terms with a difference in price levels.

The choice of deflator moves the levels and barely moves the comparison. The same plant costs 2.18 times as much under the Latvian rate on this measure, 2.18 at a 2 per cent inflation assumption, 1.98 on each country's own 2018 to 2019 inflation, and 2.10 if the nominal rates are used without conversion. Figure 6 uses stated real rates throughout and involves no conversion.

Data and replication

All tables and figures are generated programmatically from the underlying data. Replication materials, including the data-processing and analysis code, the supporting datasets, the robustness checks and the documentation of the historical sources, are available in the online replication package. Each reported output can be traced to its processed dataset and, where applicable, to the source extract behind it. The historical financing evidence is documented separately, including how the archival documents were identified and read and what the available archive cannot reach.

Replication materials will be made available with the final version of the paper.

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